



Climate disaster risks, response capacities and sovereign bond yield spreads

Wenpeng Chen^{a,1}, Jiaqi Wang^{b,1}, Chao Zhao^{c,*}

^a School of Finance, Guangdong University of Foreign Studies, Institute of Financial Openness and Asset Management, Guangdong 510000, China

^b School of Finance and Economics, Guangdong Polytechnic Normal University, Guangdong 510000, China

^c School of Economics, Xiamen University, Fujian 361005, China

ARTICLE INFO

Keywords:

Asset pricing

Rare disaster

Extreme heat

Sovereign bond yield spreads

ABSTRACT

From the perspective of extreme heat, this study integrates theoretical model expansion and empirical testing to investigate how climate disasters influence sovereign bond yield spreads and the effectiveness of climate response capabilities. Specifically, it first incorporates extreme heat into the rare disaster theoretical framework and extends the asset pricing model to provide theoretical evidence on how extreme heat influences sovereign bond yield spreads. Next, by utilizing an unbalanced panel of macroeconomic data for 49 countries over the period 1995–2018, this research applies the Least Squares Dummy Variable Corrected (LSDVC) approach for empirical tests concerning the effects, transmission channels, and response measures associated with extreme heat. The findings reveal that: (1) Extreme heat increases sovereign bond yield spreads across various nations, with the yield spread increase being more pronounced for developing countries; (2) The primary transmission channels through which extreme heat affects sovereign bond yield spreads include damage to agricultural production and repercussions on social stability, both of which raise the risk of sovereign default; (3) Enhancements in climate response capacity can effectively mitigate the increase in sovereign bond yield spreads caused by extreme heat. In response to extreme heat shocks, the business environment and social readiness of an economy all have significant alleviating effects. This study enriches the research on climate finance by concentrating on extreme heat and identifies the key sectors for responses to such climate events.

1. Introduction

The drastic effects of global climate warming may represent the most significant challenge facing humanity in this century. An analysis by NASA of global surface temperatures reveals that 19 of the 20 hottest years on record since 1880 occurred within the first 21 years of the new century, with both 2016 and 2020 being tied as the warmest years recorded.² Climate warming accelerates the melting of ice and snow at the polar regions, resulting in rising sea levels and increasing incidence of natural disasters like heatwaves, droughts, wildfires, floods, and typhoons. According to a report co-published by the World Wildlife Fund and other organizations (Price et al., 2018), if global temperatures rise by 4.5 °C in the future, between 40% to 50% of Earth's species could face extinction. Furthermore, climate warming is projected to incur enormous losses to human assets. A quantitative analysis conducted by The Economist Intelligence Unit (Economist Intelligence Unit EIU, 2015)

regarding that a global temperature increase of 6 °C could lead to a decline in total global asset value by around 30% due to the effects of climate change risks on investments and asset values.³

As global climate warming intensifies, extreme weather poses significant challenges to human survival and economic development, severely threatening socio-economic progress and financial systems. Fluctuations in asset prices can reflect the economic harm caused by climate change. So far, research on the risks linked to climate disasters and their impact on asset price volatility has predominantly focused on the stock market (Bansal, Kiku, & Ochoa, 2019; Bonato, Cepni, Gupta, & Pierdzioch, 2023; Karydas & Xepapadeas, 2022), the real estate market (Giglio, Maggiori, Rao, Stroebel, & Weber, 2021), and the municipal bond market (Painter, 2020), with limited attention paid to the sovereign bond market (Klusak, Agarwala, Burke, Kraemer, & Mohaddes, 2023). However, from an international perspective, sovereign bonds serve as a key channel for cross-border financing for governments and

* Corresponding author.

E-mail address: officialzchao@126.com (C. Zhao).

¹ Wenpeng Chen, Jiaqi Wang and Chao Zhao contributed equally to this paper.

² Sources: <https://climate.nasa.gov/vital-signs/global-temperature/>.

³ Sources: <https://www.aviva.com/.../socialpurpose/pdfs/thoughtleadership/EIU-cost-of-inaction.pdf>.

are an important component of international investors' and global capital's asset allocation strategies. Domestically, sovereign bond yield spreads often serve as the benchmark interest rate in financial markets, influencing the pricing of various asset classes. Recently, there has been an increase in sovereign bond defaults alongside a rising trend in default risk—potentially linked to the disparate impacts of climate disasters across different economies due to global climate warming.⁴ Investigating the effects of climate disasters and response capabilities on the sovereign bond market holds considerable theoretical and practical importance for promoting the openness of bond markets, ensuring efficient bond financing channels, and attracting as well as leveraging international capital across diverse economies.

Recent studies examining the connection between climate change and asset pricing primarily focus on two aspects: physical climate risk and transition risk. A small number of scholars have investigated the risk premium associated with physical risks (Baldauf, Garlappi, & Yannelis, 2020; Huynh, Nguyen, & Truong, 2020), while the majority of studies have concentrated on the risk premia related to transition risks, such as carbon emissions (Bauer, Huber, Rudebusch, & Wilms, 2022; Bolton & Kacperczyk, 2021; Sankar, Nag, Chakrabarty, & Basu, 2024) and ESG premia (Pástor, Stambaugh, & Taylor, 2021). In the sovereign bond market, there has been comparatively less focus on climate risk premia, with greater emphasis placed on economic risk premia such as government fiscal balances (Gruber & Kamin, 2012; Peppel-Srebrny, 2021), macroeconomic conditions (Andrade, Ekponon, & Jeanneret, 2023), countries' environmental, social, and governance (ESG) performance (Capelle-Blancard, Crifo, Diaye, Oueghli, & Scholtens, 2019), as well as the effects of the COVID-19 pandemic (Corradin & Schwaab, 2023). Kling, Lo, Murinde, and Volz (2018) examine the connection between climate vulnerability and sovereign bond yields, finding that an economy's vulnerability to climate-related risk results in elevated long-term sovereign bond yields, while some adaptation measures can effectively reduce these long-term yields. Klusak et al. (2023) further show that climate risks are likely to increase sovereign borrowing costs and generate heterogeneous effects across countries.

Within this broader literature, a growing body of work explicitly links natural disasters and climate-related shocks to sovereign risk and market pricing. Klomp (2015) shows that major natural disasters raise sovereign default premia in emerging markets, providing early evidence that disaster risk is priced in sovereign credit markets. Di Tommaso, Foglia, and Pacelli (2023) document that natural disasters lead to higher sovereign CDS spreads in Europe and generate contagion effects across countries, while Yang and Hamori (2023) model the global sovereign credit network under climate change and highlight that climate-related shocks strengthen tail dependence in sovereign credit risk and amplify cross-country spillovers. From a structural perspective, Mallucci (2022) incorporates natural disaster risk into a quantitative sovereign default model and shows that climate change, by increasing the frequency and severity of disasters, tightens governments' borrowing conditions, whereas disaster clauses can partially alleviate this constraint. Diarra & Jaber (2022) provide panel evidence that temperature anomalies are priced in sovereign CDS spreads, partly through their impact on macroeconomic performance and fiscal positions, thereby establishing a direct empirical link between climate conditions and sovereign default risk.

Although research on the effects of climate-related risks on sovereign bond yields is relatively scarce, many studies have confirmed the economic damage caused by rising temperatures. Research shows a negative correlation between temperature and overall output (Dell, Jones, & Olken, 2012), as temperature affects nearly all chemical, biological, and ecological processes. The adverse effects of high temperatures are

mainly reflected in two areas: first, temperature serves as a critical input in agriculture, with prolonged high temperatures significantly diminishing agricultural output (Burke & Emerick, 2016; Chen & Gong, 2021); second, rising temperatures increase impulsive and aggressive emotions among individuals (Tiihonen et al., 1997), which leads to a rise in violent crime and a decline in social stability (Heilmann, Kahn, & Tang, 2021; Jacob, Lefgren, & Moretti, 2007). Additionally, high temperatures affect mortality rates (Deschenes & Moretti, 2009), productivity (Somanathan, Somanathan, Sudarshan, & Tewari, 2021), and labor supply (Graff Zivin & Neidell, 2014), while also harming corporate production and operations (Chen, Chen, Wang, & Zhao, 2023), with repercussions for both upstream and downstream firms within supply chains (Pankratz & Schiller, 2024). Relatedly, Klomp (2014) finds that natural disasters undermine financial stability by increasing the probability of bank distress across countries, and Noth and Schüwer (2023) show that natural disasters weaken bank stability in the U.S. financial system by raising default risk, non-performing loans, and foreclosure rates, underscoring the importance of financial-sector channels in transmitting climate and disaster risks. Barro (2015) integrates environmental disasters into asset pricing models as rare disasters, while Bansal et al. (2019) extend this rare-disaster framework by including temperature anomalies; both studies demonstrate that stock returns respond to climate disasters. These studies together suggest that climate disaster risks may also impact sovereign bond yield spreads across different countries, yet they seldom focus specifically on extreme heat as a distinct physical risk or systematically examine the moderating role of climate response capacity.

Given that climate disasters are mostly caused by global climate warming, this study investigates how climate disasters affect sovereign bond yield spreads, focusing on extreme heat and the moderating influence of climate response capacity. Specifically, this study incorporates extreme heat into asset pricing models, extending the theoretical frameworks of Barro (2006) and Bansal et al. (2019). It is assumed that economic output is exposed to the increasing rare disaster risks associated with more frequent extreme heat events, which may trigger sovereign bond defaults and consequently affect sovereign bond yield spreads. Subsequently, we use an unbalanced panel dataset on extreme heat and 10-year sovereign bond yields for 49 countries over 1995–2018 and apply methods such as the Least Squares Dummy Variable Corrected (LSDVC) estimator to examine how extreme heat affects sovereign bond yield spreads and their transmission channels, as well as whether climate response capacity—along the dimensions of climate vulnerability and climate readiness—can mitigate these adverse effects. The findings indicate that extreme heat correlates with an increase in sovereign bond yield spreads, particularly within developing nations. The primary mechanisms underlying this positive effect include agriculture and social stability, and the conclusion remains valid even for middle- and high-income countries. Furthermore, climate response capacity can significantly mitigate the negative effects of extreme heat on sovereign bond yield spreads, with the food, health, and housing sectors playing a dominant role in extreme heat vulnerability.

Compared to prior studies, this study offers the following contributions: First, from the perspective of long-term climate risk, we explicitly connect global warming-induced extreme heat to climate disasters and embed this linkage into a rare-disaster asset pricing framework. Existing rare-disaster models largely focus on productivity shocks or financial market crashes. Building on Barro (2006, 2015) and Bansal et al. (2019), we extend this line of research by treating extreme heat as a driver of rare macroeconomic disasters and formally deriving how a rising frequency of extreme heat events affects the probability and severity of these disasters, the likelihood of sovereign default, and the sovereign bond default risk premium. In doing so, we move beyond reduced-form correlations between climate variables and spreads and provide a structural mechanism that translates long-run climate warming into sovereign bond yield spreads. This extends rare-disaster-based asset pricing theory from equity markets to sovereign bonds and turns the

⁴ According to the 2020 report by Standard & Poor's Financial Services, there were 20 sovereign debt defaults between 1999 and 2019, including defaults by the developed economy of Greece in January and May of 2012.

long-run climate threat into a concrete, model-based prediction for sovereign borrowing costs.

Second, this study enriches the empirical literature on climate disasters and sovereign credit markets by focusing on extreme heat and combining theoretical model expansion with systematic panel evidence. While many studies examine the pricing of climate risks in other asset classes, such as stocks (Bansal et al., 2019; Bonato et al., 2023; Karydas & Xepapadeas, 2022), real estate (Giglio et al., 2021) and municipal bonds (Painter, 2020), and recent work documents how natural disasters and climate conditions affect sovereign risk premia and CDS spreads (Di Tommaso et al., 2023; Diarra & Jaber, 2022; Klomp, 2015; Mallucci, 2022; Yang & Hamori, 2023), most of these studies rely on broad disaster indices or temperature anomalies. In contrast, we focus on extreme heat as a specific and increasingly salient physical risk by using the Warm Spell Duration Index (WSDI) to capture persistent extreme high-temperature events. Drawing on an unbalanced panel dataset on extreme heat and 10-year sovereign bond yield spreads for 49 countries over 1995–2018, and applying the Least Squares Dummy Variable Corrected (LSDVC) estimator, we provide systematic evidence that extreme heat is priced in sovereign bond yield spreads after controlling for standard macroeconomic, fiscal, and global financial determinants. Our work complements and extends the insights of Hong, Li, and Xu (2019), Pankratz, Bauer, and Derwall (2023) and Chen et al. (2023) by showing how physical climate shocks translate into sovereign borrowing costs rather than equity or firm-level outcomes.

Third, this study deepens the understanding of the economic costs of climate change and the strategies for mitigating these impacts by identifying concrete transmission channels and the role of climate response capacity. Existing research shows that climate vulnerability and adaptation policies shape long-term sovereign borrowing costs (Kling et al., 2018), and that natural disasters weaken bank stability and financial resilience (Klomp, 2014; Noth & Schüwer, 2023). Building on these insights, we establish a direct connection between extreme heat and sovereign bond yield spreads and show that the primary channels through which extreme heat affects spreads are damage to agricultural production and deterioration in social stability. These sectoral and social mechanisms are crucial drivers behind rising sovereign bond yield spreads and complement the macroeconomic and fiscal channels emphasized by Mallucci (2022) and Diarra & Jaber (2022). Furthermore, using detailed indicators of climate vulnerability and climate readiness, we show that climate response capacity can significantly mitigate the adverse effects of extreme heat on sovereign bond yield spreads, and we identify key sectors—food security, healthcare systems, living conditions, business environment, and social readiness—that are especially important for buffering sovereign credit markets against extreme heat. This provides policymakers with targeted guidance on how to stabilize sovereign financing conditions in the face of intensifying physical climate risks.

1.1. The remainder of this study is organized as follows

Section 2 introduces the theoretical model; Section 3 outlines the data and variables; Section 4 provides empirical evidence and transmission channel analysis; Section 5 discusses the effectiveness of climate response capacity; and Section 6 concludes with policy recommendations.

2. Theoretical model

2.1. Model assumptions

This study develops a theoretical model grounded in Lucas Jr (1978) representative agent framework, assuming that production is exogenous and stochastic, and introduces climate disasters on the basis of Barro (2006). Specifically, the economy is assumed to be closed, with all output devoted to consumption, that is the output in period t , denoted as

A_t , is equal to consumption C_t . In period t , there are two categories of assets available in the market. The first type is equity, priced at $P_{e,t}$, which entitles the holder to the company's output A_{t+1} in period $t + 1$. The other type is sovereign bonds, priced at $P_{b,t}$, which, in the absence of default, provide $B_{t+1} = 1$ units of consumption goods and has a yield of $R_{b,t+1}$.

It is assumed that the representative consumer possesses a power utility function:

$$U_t = E_t \sum_{i=0}^{\infty} \left[e^{-\delta i} \frac{C_{t+i}^{1-\theta}}{1-\theta} \right]$$

where, δ represents the time preference parameter and θ denotes the coefficient of relative risk aversion, with $\delta > 0$ and $\theta > 0$. Assuming that output follows a random walk process with a drift term:

$$\ln(A_{t+1}) = \ln(A_t) + \gamma + u_{t+1} + V_{t+1}$$

where γ captures the long-term growth rate of consumption, and $u_{t+1} \sim N(0, \sigma_u^2)$ represents short-term growth shocks. Given that output may be influenced by disasters, we introduce climate disasters $V_{t+1} = N_{t+1} \log(1-d)$ into the output growth rate. Here, d denotes the proportion of losses caused by the disaster, with $0 \leq d \leq 1$, and N_{t+1} is the number of climate disaster occurrences. Similar to the setup in Bansal et al. (2019), we assume $N_{t+1} \sim P(\lambda_t)$ and $\lambda_t = l_0 + l_1 WSDI_t$, where $WSDI_t$ represents extreme heat (characterized as the number of days when the maximum temperature surpasses the 90th percentile of the reference period for more than six consecutive days), $l_0 > 0$ refers to rare disasters that are triggered by various external factors, and $l_1 > 0$ represents the sensitivity of the intensity of climate disasters to extreme heat.⁵ This assumption aligns with economic realities, as discussed earlier, where extreme heat damages agriculture and social stability, and an uptick in the occurrence of extreme heat events can result in severe economic consequences. On the other hand, due to incomplete information, investors often do not know whether the disaster being experienced is temporary or persistent (Ghaderi, Kilic, & Seo, 2022), and thus they need to adjust their assessments of the probability of climate disaster occurrence grounded in the frequency of extreme heat events.

Given the number of disaster occurrences, it is assumed that the default probability of sovereign bonds is $P_{t+1} = 1 - e^{-N_{t+1}}$. If default occurs, the bondholder receives a real value of $(1-b)$ units of consumption goods at maturity. If no default occurs, the bondholder receives a real value of 1 unit of consumption goods at maturity, with a probability of $1 - P_{t+1} = e^{-N_{t+1}}$. Unlike Barro (2006) who assumes a fixed default probability, this study assumes that the default probability is time-varying and increases with the number of climate disaster occurrences. As the frequency of extreme heat events rises, so does the expected number of climate disasters and the probability of sovereign bond defaults. In other words, the probability of sovereign bond default is an increasing function of extreme heat frequency. In Appendix B, this study provides the joint probability density function of $(B_{b,t+1}, N_{t+1})$, and the marginal probability density function of $B_{b,t+1}$.

2.2. Model solution

Following Barro (2006), we posit that rare disasters inflict equivalent proportional damage on both output and the real value of sovereign bonds, that is $d = b$. The utility-maximizing Euler equation is given by:

⁵ Unlike Barro's (2006) assumption of a fixed disaster occurrence probability, this study assumes that the probability of rare disasters is time-varying. In contrast to Bansal et al. (2019), this study posits that the occurrence of rare disasters is directly related to the frequency of extreme heat events. The more frequent the extreme heat events, the higher the expected number of rare disasters.

$$P_{b,t} = E_t \left[e^{-\delta} \left(\frac{C_{t+1}}{C_t} \right)^{-\theta} B_{b,t+1} \right]$$

Based on the joint probability distribution of $B_{b,t+1}$ and N_{t+1} , $\ln(P_{b,t})$ can be derived (for the detailed derivation process, see Appendix B):

$$\ln(P_{b,t}) = -\delta - \theta\gamma + \frac{1}{2}\theta^2\sigma_u^2 + \ln \left[(1-d)e^{\lambda_t[(1-d)^{-\theta}-1]} + de^{\lambda_t \left[\frac{1}{e^{(1-d)^{-\theta}-1}} \right]} \right]$$

Based on the marginal probability distribution of $B_{b,t+1}$, the expected yield of sovereign bonds, $\ln E_t(R_{b,t+1})$, can be calculated (for the detailed derivation process, see Appendix B):

$$\begin{aligned} \ln E_t(R_{b,t+1}) &= \ln E_t \left(\frac{B_{b,t+1}}{P_{b,t}} \right) = \ln E_t(B_{b,t+1}) - \ln(P_{b,t}) \\ &\approx d(l_0 + l_1 WSDI_t) \frac{e-1}{e} \left[\frac{1}{(1-d)^\theta} - 1 \right] + \ln E_t(R_{b,t+1}^0) \end{aligned}$$

where $\ln E_t(R_{b,t+1}^0) = -\ln\delta + \theta\gamma - \frac{\theta^2}{2}\sigma_u^2 - (l_0 + l_1 WSDI_t) \left[(1-d)^{-\theta} - 1 \right]$ denotes the expected yield of sovereign bonds when regarded as risk-free assets, i.e., the expected yield in the absence of defaults attributable to climate disasters. While $RC = d(l_0 + l_1 WSDI_t) \frac{e-1}{e} \left[\frac{1}{(1-d)^\theta} - 1 \right] > 0$ is the compensation for sovereign bond default risk.

2.3. Model implication

The marginal impact of extreme heat on the sovereign bond default risk premium:

$$\frac{\partial RC}{\partial WSDI_t} = dl_1 \frac{e-1}{e} \left[\frac{1}{(1-d)^\theta} - 1 \right] > 0$$

which indicates that the risk premium for sovereign bond defaults due to extreme heat is positive, implying that a rise in extreme heat will result in elevated sovereign bond yield spreads. Furthermore, Eq. (7) suggests that the marginal impact of extreme heat depends on the sensitivity of rare disasters to extreme heat, denoted as l_1 :

$$\frac{\partial^2 RC}{\partial WSDI_t \partial l_1} = d \frac{e-1}{e} \left[\frac{1}{(1-d)^\theta} - 1 \right] > 0$$

It is important to note that $l_1 = 0$ means that extreme heat does not influence the probability of rare disasters occurring, thus the degree of correlation between extreme heat and rare disasters depends on the value of l_1 . This equation also indicates that measures aimed at reducing the correlation between extreme heat and rare disasters can effectively mitigate the effect of extreme heat on sovereign bond yield spreads. For instance, if an economy's climate response capacity can alleviate the negative effects of extreme heat, then a higher climate response capacity will correspond to a diminished negative effect of extreme heat on its economy and society, as well as a reduced marginal effect.

Finally, the marginal impact of other factors (l_0) that induce rare disasters is:

$$\frac{\partial RC}{\partial l_0} = d \frac{e-1}{e} \left[\frac{1}{(1-d)^\theta} - 1 \right] > 0$$

When the marginal impact of other factors is positive, even if extreme heat does not directly induce rare disasters ($l_1 = 0$), extreme heat may still have a positive correlation with sovereign bond yield spreads if it is positively correlated with other factors that lead to $l_0 > 0$. For example, rising sea levels or droughts could trigger climate disasters, and extreme heat is often positively correlated with these two physical

risks. Consequently, even when rare disasters are not directly caused by extreme heat, a positive correlation between extreme heat and sovereign bond yield spreads may still be observed. The proposition of the theoretical model⁶ do not exclude this confounding factor, thus this study will develop relevant empirical models to investigate it in the empirical section. Based on the aforementioned theoretical model, this study derives the subsequent propositions:

Proposition 1:

The rising occurrence of extreme heat events correlates with an increased risk of sovereign bond default, as indicated by wider sovereign bond yield spreads.

Proposition 2:

An economy's climate response capacity can mitigate the adverse impact of extreme heat on sovereign bond yield spreads.

3. Data and methodology

3.1. Climate data and independent variables

The extreme weather data is sourced from the Climdex database,⁶ which is sponsored by the Australian Research Council. This database provides three data options: the "HadEX2" dataset created by Donat et al. (2013a), the "GHCNDEX" dataset created by Donat et al. (2013b), and the "HadEX3" dataset compiled by Dunn et al. (2020). This study employs the most statistically refined dataset, "HadEX3", which includes data from 7000 global temperature stations and 17,000 precipitation stations spanning from 1901 to 2018. Given the incompleteness of earlier data, only the sample from 1951 to 2018 are retained, excluding countries with less than 80% valid data coverage. Consequently, this study constructs an unbalanced panel climate dataset covering 160 countries over a span of 68 years, comprising a total of 10,737 observations. The indicators representing extreme temperatures in this dataset include warm days ($TX90p$), warm nights ($TN90p$), cold days ($TX10p$), cold nights ($TN10p$), the Warm Spell Duration Index ($WSDI$), and the Cold Spell Duration Index ($CSDI$), among others, and their definitions can be found in Table 1.

In this paper, we use the Warm Spell Duration Index ($WSDI$) as our main measure of extreme heat. Compared with the simple frequency of hot days ($TX90p$), this duration-based index is better suited to our

Table 1
Explanation of extreme temperature indicators.

Abbreviation	Indicator Name	Description
$WSDI$	Warm Spell Duration Index	Number of days with daily maximum temperature above the 90th percentile of the reference period for more than 6 consecutive days.
$CSDI$	Cold Spell Duration Index	Number of days with daily maximum temperature below the 10th percentile of the reference period for more than 6 consecutive days.
$TX90P$	Warm Days	Percentage of days where the daily maximum temperature exceeds the 90th percentile of the reference period.
$TN90P$	Warm Nights	Percentage of days where the daily minimum temperature exceeds the 90th percentile of the reference period.
$TN10P$	Cold Nights	Percentage of days where the daily minimum temperature is below the 10th percentile of the reference period.
$TX10P$	Cold Days	Percentage of days where the daily maximum temperature is below the 10th percentile of the reference period.

⁶ Sources: <https://www.climdex.org/>.

research question for both economic and conceptual reasons. From an agronomic and production perspective, cumulative exposure to consecutive hot days is considerably more damaging to crops and outdoor work than isolated hot days, particularly when heat waves coincide with critical stages of plant growth or peak agricultural activity. Similarly, from a health and social perspective, heat waves rather than sporadic hot days are more strongly associated with excess mortality, hospital admissions, and heightened social tension, because there is little time for recovery when high temperatures persist without cooling intervals. These features make *WSDI* a more natural proxy for the type of persistent, high-impact heat events that can trigger rare-disaster-like macroeconomic and fiscal stress in our theoretical framework, and thus more directly linked to sovereign default risk and bond yield spreads than indicators that only count the overall number of hot days.

The calculation of these six indicators is based on the reference period from 1961 to 1990. In this study, the arithmetic mean of these six indicators is computed at the national level by averaging across grid points based on latitude and longitude, resulting in six national-level extreme temperature indicators. Fig. 1 illustrates the distribution of average warm days across countries for the period from 1995 to 2018. Among the sampled countries, nearly all have experienced average warm days exceeding 10%, with those in Eurasian collectively exceeding 15%. After calculating the arithmetic mean of the national-level extreme temperature indicators, the global-level extreme temperature indicators are derived. Fig. 2 illustrates the temporal trends of global extreme temperature indicators. The Warm Spell Duration Index (*WSDI*) and warm days (*TX90p*), which represent extreme heat, show a mild upward trend over the sample period, while the Cold Spell Duration Index and cold days, representing extreme cold, show a mild downward trend. By 2018, the global average percentage of warm days had nearly reached 20%, indicating that the occurrence of extreme heat events has doubled in the last 38 years.

3.2. Dependent variable and control variables

The primary dependent variable examined in this study is the sovereign bond yield spreads, with data sourced from the [Investing.com](#) database. Consistent with [Jahjah, Wei, and Yue \(2013\)](#) and [Cevik and Jalles \(2022\)](#), we use the yield on the U.S. 10-year Treasury bond as a proxy for the risk-free rate, denoted as $R_{b,t+1}^0$ in Eq. (5). This choice, first, imposes a common benchmark risk-free rate across countries and, given the depth and stability of the U.S. Treasury market, the 10-year yield provides an appropriate reference. Second, it mitigates data limitations arising from the lack of reliable domestic risk-free rate series for some countries. The yield spread for each nation is determined by subtracting the U.S. Treasury yield from country's 10-year sovereign bond yield. For countries such as Argentina, Botswana, Pakistan, and Ukraine, which do not have 10-year sovereign bonds, the longest available maturities (7/5/3/3 years, respectively) are used as substitutes.⁷ The decision to focus on 10-year bond spreads is based on two key considerations. First, climate change risks are perceived primarily as long-term threats, and extreme heat is more likely to trigger climate disasters over the long term, making long-term bonds more apt to incorporate this risk into their pricing. Second, unlike short-term bonds, which are more easily influenced by monetary policy, the default risk of long-term bonds warrants greater attention.

Referring to previous literature ([Capelle-Blancard et al., 2019](#); [Gruber & Kamin, 2012](#); [Jahjah et al., 2013](#); [Peppel-Srebrny, 2021](#)), this study controls for a series of macroeconomic variables in the model: short-term interest rate (*SIR*), GDP growth rate (ΔGDP), current account balance as a percentage of GDP (*CA*), inflation rate (*Inflation*),

unemployment rate (*Unemployment*), credit rating (*Credit*), fiscal deficit as a percentage of GDP (*Deficit*), government debt as a percentage of GDP (*Debt*). In addition, we include the VIX index and the CRB index to control for the potential influence of global risk appetite and commodity prices on sovereign bond yield spreads. Finally, we also control for the maximum temperature (*TXx*). The GDP growth rate, current account balance as a percentage of GDP, inflation rate, and fiscal deficit as a percentage of GDP are sourced from the IMF World Economic Outlook (WEO) dataset, the unemployment rate is sourced from the World Bank Open Database, and the credit rating is sourced from the Wind database. This study employs Standard & Poor's long-term sovereign credit ratings, assigning a value of 1 to the lowest rating (C), 2 to the second-lowest rating (CC), and so forth, with the top rating (AAA) is given a value of 21. The definitions of the dependent variable and control variables are presented in Table 2.

The short-term interest rate is obtained from both the OECD database and the CEIC database. To minimize the sample loss from developing countries, the primary source for the short-term interest rate is real data from the OECD database. For missing values, we draw the corresponding time-series data from the CEIC database, a standardized macroeconomic database covering over 190 economies; remaining gaps are then filled using OECD forecast data.

For clarity, the CEIC database refers to the China Economic Information Center database, operated by CEIC Data Company Ltd. It is a leading global macroeconomic database that provides high-quality, cleaned, and standardized time-series data covering macroeconomic, industry, and financial indicators for more than 190 countries and regions, with particularly strong coverage of China and other emerging markets.

Ultimately, 13 observations were replaced with OECD forecast data, resulting in a negligible overall impact.

3.3. Descriptive statistics and stationarity tests

After the integration of climate data with sovereign bond yield data, this study has constructed an unbalanced panel dataset encompassing 49 countries from 1995 to 2018, with a total of 757 observations.⁸ Table 3 displays the descriptive statistics for the main variables, and the country distribution of the sample is detailed in Appendix Table A1..

To avoid spurious regression, we examine stationarity using the Fisher-type panel unit-root test with a drift term ([Choi, 2001](#); [Maddala & Wu, 1999](#)). Our dataset is an unbalanced panel with heterogeneous time spans across units; the Fisher approach aggregates *p*-values from unit-specific ADF regressions, allowing unit-specific lag lengths and unequal *T*, so that no observations need to be discarded to balance the panel. This is preferable to applying conventional ADF uniformly, which would require dropping units/years or imposing a common lag order and hence reduce power for short series. We evaluate lag orders 1–5. Table 4 reports the results: for all variables the unit-root null is rejected at the 1% level, so we treat the variables as stationary in subsequent regressions.

3.4. Methodology

Considering the high autocorrelation of sovereign bond yields, this study constructs a dynamic panel fixed-effects model to estimate the impact of extreme temperatures on sovereign bond yield spreads. Specifically, the empirical equation in this study is as follows:

$$Spread_{it} = \beta_0 + \beta_1 Spread_{it-1} + \beta_2 climate_{it} + \epsilon' z_{it} + A_i + B_t + \epsilon_{it}$$

⁷ In the robustness check, we estimate using a subsample that excludes these countries, and the results do not show significant changes.

⁸ The beginning of the sample period is constrained by the availability of sovereign bond yield data, while the ending is restricted by the accessibility of climate data.

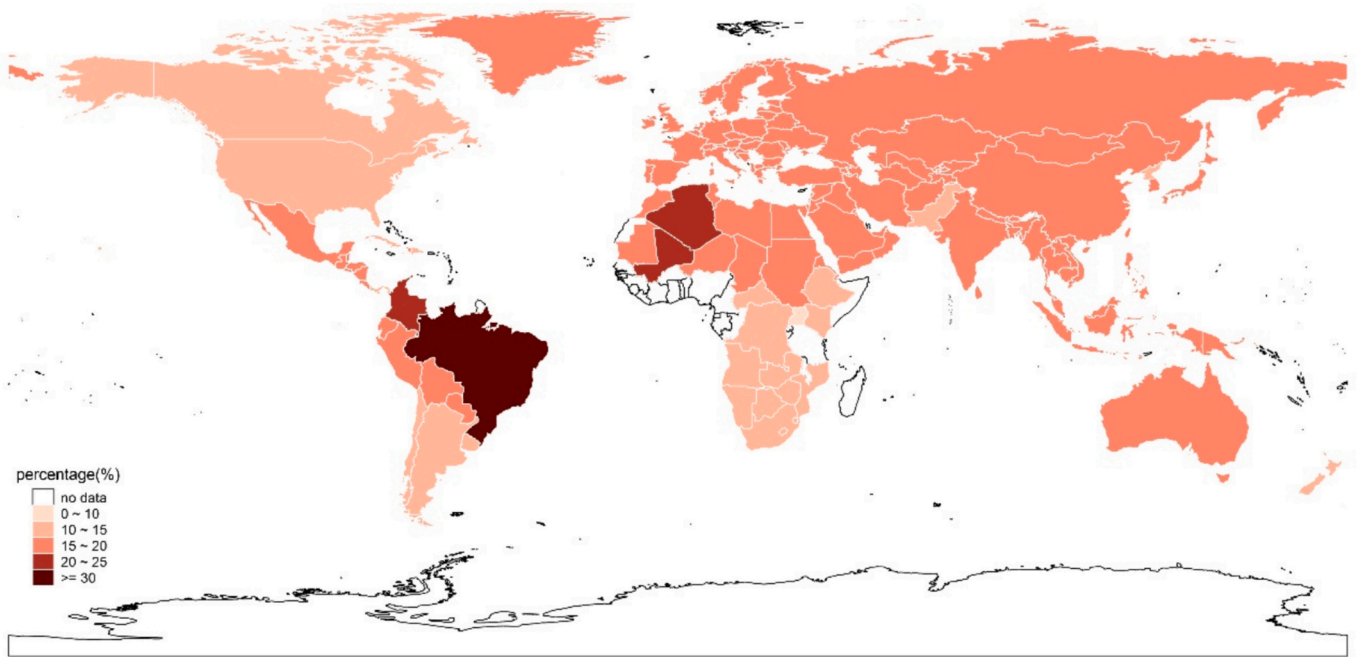


Fig. 1. Global distribution of warm day frequency (1995–2018 average).

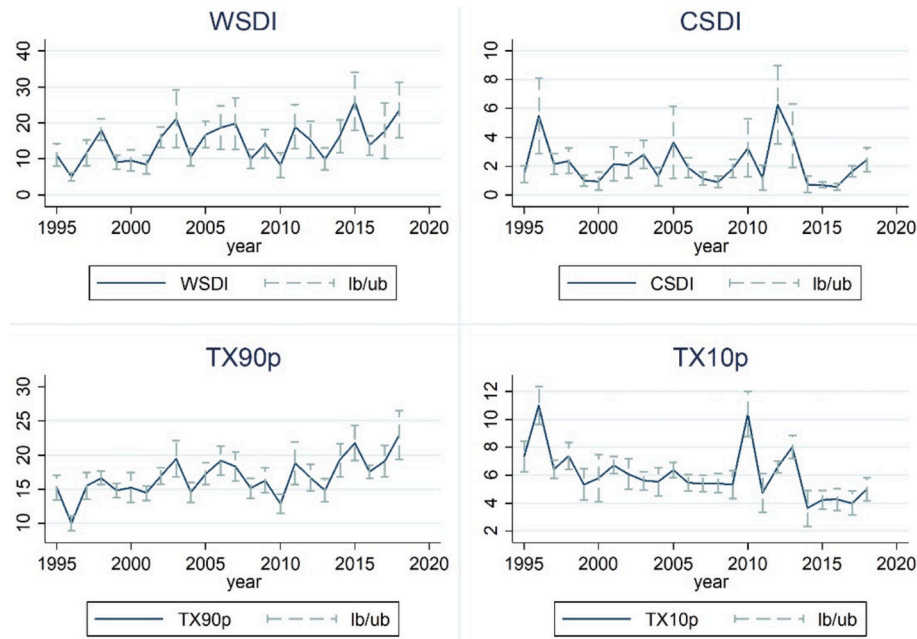


Fig. 2. Time trends of extreme temperatures.

where, i stands for country, t indicates year. The control variables z_{it} include macroeconomic variables, VIX Index, CRB Index and maximum temperature, and ε_{it} represents the error term. A_i represents country-level fixed effects, which control for country-specific heterogeneity that does not change over time, such as the natural environment, relatively stable national creditworthiness, and economic strength. B_t represents year fixed effects, which control for the time-specific effects on sovereign bond yield spreads, such as the global downward trend in bond yields and major financial crises.

Direct estimation of Eq. (9) may result in endogeneity issues, as the presence of unobserved individual effects A_i will cause an upward bias in the OLS estimate of the coefficient for the lagged dependent variable, β_1 ,

while the fixed-effects estimate of β_1 will exhibit a downward biased (Nickell, 1981). Therefore, the consistent estimator of β_1 will fall between the mixed OLS and fixed-effects estimates. Given that this study's data consists of a 24-year long panel, the LSDVC method is chosen to estimate Eq. (9) to address this endogeneity issue. In the appendix, we also present the model estimation results using both system GMM and difference GMM methods to confirm that the findings remain robust across various estimation techniques.

Furthermore, although the Warm Spell Duration Index is a natural phenomenon and is nearly completely exogenous, eliminating concerns about the endogeneity of the core explanatory variable, this does not imply that the model's estimation results are free from endogeneity

Table 2

Dependent variable and control variables.

Abbreviation	Indicator Name	Indicator Description
<i>Spread</i>	Yield Spread	10-year sovereign bond yield minus the U. S. 10-year Treasury yield
<i>SIR</i>	Short-term Interest Rate	3-month interbank lending rate or sovereign bond yield
ΔGDP	GDP Growth Rate	Annual GDP growth rate
<i>Inflation</i>	Inflation	Inflation rate measured by CPI
<i>CA</i>	Current Account Balance	Current account balance as a percentage of GDP
<i>Unemployment</i>	Unemployment Rate	Number of unemployed / labor force
<i>Credit</i>	Sovereign Credit Rating	Based on S&P ratings, with values from 1 to 21 corresponding to C to AAA
<i>Deficit</i>	Fiscal Deficit	Fiscal deficit as a percentage of GDP
<i>Debt</i>	Government Debt	Government debt as a percentage of GDP
<i>Precip</i>	Precipitation	Total annual precipitation
<i>TXx</i>	Maximum Temperature	the highest recorded daily maximum temperature
<i>Lat</i>	Latitude	Latitude of the country's central point
<i>VIX</i>	VIX Index	CBOE Volatility Index
<i>CRB</i>	CRB Index	Thomson Reuters Core Commodity CRB Index

Table 3

Descriptive statistics.

Variable		Obs	Mean	Sd	Min	Max
Economic Variables	<i>Spread</i>	757	2.100	4.070	−2.980	22.67
	<i>SIR</i>	750	4.100	3.680	−0.780	26.61
	ΔGDP	757	2.950	2.960	−10.15	25.31
	<i>Inflation</i>	755	3	3.110	−2.540	29.78
	<i>CA</i>	757	0.030	5.270	−23.91	16.30
	<i>Unemployment</i>	757	7.840	5.440	0.210	33.29
	<i>Credit</i>	756	16.72	4.230	2	21
	<i>Deficit</i>	755	2.410	4.120	−18.63	32.03
	<i>Debt</i>	755	63.24	37.94	4.920	233.5
	<i>WSDI</i>	757	16.78	10.33	0	47.55
Climate Variables	<i>CSDI</i>	741	2.330	2.940	0	15.79
	<i>TX90p</i>	751	17.58	4.400	6.310	32.97
	<i>TN90p</i>	757	18.75	5.180	8.040	39.76
	<i>TX10p</i>	751	6.440	2.290	1.950	12.78
	<i>TN10p</i>	757	6.190	2.140	0.810	12.67
	<i>Precip</i>	754	1020	662.9	0	4731
	<i>TXx</i>	757	34.50	4.440	18.53	44.38
	<i>Lat</i>	757	39.30	14.70	0.790	64.96

Table 4

Fisher unit root test.

Variable	Lag 1	Lag 2	Lag 3	Lag 4	Lag 5
<i>spread</i>	0.000	0.000	0.000	0.000	0.000
<i>WSDI</i>	0.000	0.000	0.000	0.000	0.000
<i>Interest</i>	0.000	0.000	0.000	0.005	0.000
ΔGDP	0.000	0.000	0.000	0.000	0.000
<i>Cpi</i>	0.000	0.000	0.000	0.000	0.000
<i>CA</i>	0.000	0.000	0.000	0.000	0.000
<i>Unemployment</i>	0.000	0.000	0.000	0.000	0.000
<i>Credit</i>	0.000	0.000	0.000	0.000	0.000
<i>Deficit</i>	0.000	0.000	0.000	0.000	0.000
<i>Debt</i>	0.000	0.000	0.000	0.000	0.000

Note: All tests are Fisher unit root tests with drift terms. The numbers in the table represent p-values.

issues. There may be two types of endogeneity between sovereign bond yields and fiscal decisions (Peppel-Srebrny, 2021). First, sovereign bond yields measure the government's long-term financing costs, which may have a reverse causal effect on fiscal decisions. Second, there may be simultaneous changes due to the economic cycle. To mitigate the bias caused by the endogeneity of control variables, the study introduces the lagged one-period fiscal deficit and government debt into the model to

control for reverse causality, and macroeconomic control variables are included to account for the effects of the economic cycle. Additionally, credit ratings may also be endogenous, as both sovereign bond yields and credit ratings may depend on the same factors, making it difficult to clearly identify the causal relationship between them (Larraín et al., 1997). Therefore, we only include the lagged one-period credit rating in the robustness checks.

4. Empirical test

4.1. Benchmark regression

The baseline regression results are shown in Table 5. All columns are based on the LSDVC model using initial values from the Blundell and Bond (1998) estimator, with 50 iterations of bootstrapping, and a bias precision of $o(n^{-1}T^{-2})$. The model controls for the fixed effects related to both year and country. In the overall five columns of results, the coefficient for the lagged one-period sovereign bond yield spreads ranges from 0.578 to 0.642, falling within the boundaries formed by the corresponding OLS and FE model estimates, indicating the correctness of the estimation method.

Specifically, in Column (1), the independent variables include only the lagged spread (*L.Spread*) and the *WSDI*. The estimated coefficient for the Warm Spell Duration Index is 0.014 and is significant at the 95% confidence level. In Columns (2) to (5), the estimation equation progressively incorporates macroeconomic indicators and climate variables. The coefficient of the core explanatory variable, *WSDI*, remains stable between 0.023 and 0.028, achieving significance at least at the 95% level. In Column (5), after including the control variable for maximum temperature, the coefficient of *WSDI* increases to 0.028 and attains significance at the 99% level. This indicates that controlling for regional differences in maximum temperature enhances the effect of

Table 5

The impact of extreme heat on sovereign bond yield spreads.

Variable	(1)	(2)	(3)	(4)	(5)
<i>L.Spread</i>	0.642*** (23.79)	0.595*** (14.26)	0.583*** (16.06)	0.583*** (16.00)	0.580*** (16.09)
<i>WSDI</i>	0.024** (2.13)	0.025** (2.33)	0.024*** (2.75)	0.023*** (2.74)	0.028*** (2.71)
<i>SIR</i>		0.222*** (3.41)	0.140** (2.28)	0.147** (2.33)	0.147** (2.36)
ΔGDP			−0.133*** (−2.58)	−0.129** (−2.40)	−0.134** (−2.55)
<i>Inflation</i>			0.086* (1.87)	0.082* (1.82)	0.084* (1.82)
<i>CA</i>			0.004 (0.13)	0.010 (0.33)	0.009 (0.29)
<i>Unemployment</i>			0.051 (1.25)	0.024 (0.52)	0.015 (0.34)
<i>L.Deficit</i>				0.036 (0.86)	0.036 (0.86)
<i>L.Debt</i>				0.006 (0.86)	0.007 (1.06)
<i>VIX</i>				0.062** (2.31)	0.057** (2.15)
<i>CRB</i>				0.005 (1.62)	0.005* (1.71)
<i>TXx</i>					−0.091 (−1.07)
Country/Year FE	Yes	Yes	Yes	Yes	Yes
<i>N</i>	694	689	688	688	685
Countries	49	48	48	48	48

Note: All estimation results are based on the Blundell & Bond (1998) estimator, with 50 iterations of bootstrapping and bias precision $o(n^{-1}T^{-2})$ in the LSDVC model. *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively. Unless otherwise specified, the estimation methods in the following sections are consistent with those used in this table.

extreme heat on sovereign bond yield spreads. The estimated coefficient implies that for every extra day in the *WSDI*, there is a 0.028% rise in the sovereign bond yield spreads. With a sample mean of 16.78 days in this index, this translates to an approximate 0.47% increase in the sovereign bond yield spreads annually. Given the mean sovereign bond yield spreads of 2.10%, the impact of extreme heat is considerable..

The estimated coefficient for maximum temperature (*TXx*) lacks statistical significance, suggesting that the sovereign bond market is not affected by isolated instances of high temperatures, instead, it is affected by the prolonged and recurring extreme heat events. In addition, regarding macroeconomic variables, the estimated coefficients for short-term interest rate, GDP growth rate, and inflation exhibit relative robustness. Both short-term interest rate and inflation show a positive correlation with sovereign bond yield spreads, whereas GDP growth rate negatively influences these yields. The lagged coefficients for government debt and fiscal deficit are positive but not significant, which may be attributed to the use of lagged variables, as previous research has already demonstrated the relationship between government fiscal conditions and sovereign bond yields (Gruber & Kamin, 2012; Peppel-Srebrny, 2021).

4.2. Robust test

4.2.1. Excluding the impact of extreme cold

In the baseline regression, the study only considers the Warm Spell Duration Index as an indicator of extreme heat. In fact, this dataset also includes a closely related indicator, warm days (*TX90p*), which serves as an alternative metric for analysis. Moreover, as illustrated in Fig. 1, the trends of extreme heat and extreme cold exhibit inverse relationships. To enhance the credibility of the baseline regression results, it is essential to exclude any potential effects of extreme cold on sovereign bond yield spreads. Therefore, we further conduct empirical tests on both indicators of extreme heat and extreme cold.

The findings presented in Table 6 reveal that the estimated coefficients for both the *WSDI* and *TX90p* are significantly positive, indicating that the baseline regression outcomes are not driven by specific indicators alone. In contrast, the coefficients of three indicators representing cold temperatures—cold days (*TX10p*), cold nights (*TN10p*), and the Cold Spell Duration Index (*CSDI*)—are statistically insignificant. The results from Table 6 suggest that only extreme heat affects sovereign bond yield spreads, and despite their opposing trends, extreme cold does not impact sovereign bond yield spreads. This finding reinforces concerns that climate warming poses a threat to global economic growth. Furthermore, the estimated coefficient for warm nights (*TN90p*) is neither significant in Table 6, which aligns with the economic reality that most economic activities occur during the daylight hours, meaning extreme heat likely affects the sovereign bond market primarily through its impact on economic production.

4.2.2. Excluding the interference of other climate-related physical disasters

The physical risks associated with global warming encompass extreme heat, increasing sea levels, and drought. However, these three disasters do not exist independently; extreme heat is often accompanied by increased risks of severe drought and rising sea levels.⁹ The static analysis of the theoretical model suggests that this correlation may affect how extreme heat impacts sovereign bond yield spreads. To address concerns about this correlation, this study designs three testing methods:

First, we test whether the impact of extreme heat differs across regions with different latitudes. Physical environments vary greatly across latitudes, particularly in terms of average temperature, with different gaps between local average temperatures and the optimal temperatures

for biological survival. However, the adverse effects of drought risk and sea level rise risk do not vary by latitude. In countries located at lower latitudes, biological systems thrive in environments that closely approximate their optimal temperatures, rendering them more sensitive to temperature fluctuations and consequently more vulnerable to the effects of climate change (Deutsch et al., 2008). In these areas, the likelihood of climate disasters induced by global warming significantly escalates, suggesting that their government bond yields should correspondingly increase.

Second, we investigate whether the impact of extreme heat differs under varying sea level rise risks. If sea level rise risk serves as a primary driver for increased sovereign bond yield spreads, then one would expect a more pronounced effect from extreme heat in regions facing higher risks associated with rising sea level. This study uses two indicators provided by the Notre Dame Global Adaptation Index (ND-GAIN) to assess vulnerability to sea level rise risks: the number of people living below 5 m above sea level (LSR1) and the predicted impact of sea level rise (LSR2).¹⁰ The indicators range from [0,1], with lower values representing lower risk. According to the descriptive statistics in Table 3, the average values of LSR1 and LSR2 are 0.28 and 0.21 respectively, suggesting that the countries within our sample generally face relatively low sea level rise risks. Given that sea level rise risk shows very little variation and does not change over time, fixed effects will control for the time-invariant risk. Therefore, we use interaction terms for the test.

Lastly, we test whether precipitation significantly affects the estimated coefficient of extreme heat. To examine whether drought drives the estimation results, this study controls for annual precipitation (*Precip*) in Eq. (4). The descriptive statistics in Table 3 show that the mean annual precipitation is 1020 mm, while the median stands at 842 mm. There are 91 sample observations with annual precipitation below 500 mm (the threshold for semi-arid conditions), accounting for 13% of the total sample, indicating that the countries within this dataset generally face a relatively low risk of drought. By comparing results from models that control solely for annual precipitation against those incorporating additional control variables related to precipitation, we can discern its impact on the estimation outcomes.

To conduct this test, it is necessary to introduce interaction terms in Eq. (9):

$$\text{spread}_{it} = \beta_0 + \beta_1 \text{spread}_{it-1} + \beta_2 \text{WSDI}_{it} + \sum_{j=1}^J \gamma_j \text{WSDI}_{it} * I_{it}^j + \sum_{j=1}^J \eta_j I_{it}^j + \tau' z_{it} + A_i + B_t + \varepsilon_{it}$$

where I_{it}^j is the j -th dummy variable (with j depending on how many groups the sample is divided into). In this section, the sample is categorized into two groups, generating one dummy variable. In Section 5, the sample is further segmented into three groups, generating two dummy variables. For example, in Table 7, $I\{Lat \geq 40\}$ represents the dummy variable for latitude (*Lat*) greater than 40°. If the latitude of a country's central point exceeds 40°, $I = 1$; otherwise, $I = 0$. When the value of the dummy variable does not change over time, there is no need to include the dummy variable separately in the model, as it would be perfectly collinear with the fixed effects. For samples that satisfy $I = 0$, the effect of extreme heat on sovereign bond yield spreads is β_2 ; for samples that satisfy $I = 1$, the total effect of extreme heat on sovereign bond yield spreads is $(\beta_2 + \gamma_j)$.

The estimation results are displayed in Table 7. Columns (1) and (2) illustrate the estimation findings for testing latitude differences, and show that whether the latitude cutoff is set at 40° or 45°, the estimated coefficient for the interaction term is significantly negative, while the coefficient for the Warm Spell Duration Index is significantly positive.

⁹ When we perform a dynamic panel regression of the Warm Spell Duration Index on annual precipitation, we find a significant negative correlation between the Warm Spell Duration Index and precipitation.

¹⁰ The dataset is introduced in detail in Section 5 of this study.

Table 6
Excluding the impact of extreme cold.

Variable	(1) Extreme Heat	(2)	(3)	(4) Extreme Cold	(5)	(6)
<i>L.Spread</i>	0.577*** (15.97)	0.577*** (19.42)	0.601*** (17.76)	0.598*** (17.66)	0.578*** (15.38)	0.599*** (17.65)
<i>WSDI</i>	0.028*** (2.71)					
<i>TN90p</i>		−0.022 (−0.73)				
<i>TX90p</i>			0.055** (2.41)			
<i>CSDI</i>				−0.022 (−0.60)		
<i>TN10p</i>					0.004 (0.08)	
<i>TX10p</i>						−0.066 (−1.23)
<i>Controls</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>Country/Year FE</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	688	702	695	673	701	687
<i>Countries</i>	48	49	48	48	48	48

Note: *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively. Control variables include macroeconomic variables and climate variables, unless otherwise specified.

Table 7
Excluding the interference of other climate-related physical disasters.

Variable	(1) Extreme Heat	(2)	(3)	(4)	(5)	(6)
<i>L.Spread</i>	0.574*** (15.96)	0.580*** (16.04)	0.578*** (16.01)	0.577*** (16.28)	0.641*** (17.37)	0.574*** (16.87)
<i>WSDI</i>	0.046*** (3.20)	0.041*** (3.13)	0.032** (2.10)	0.030** (2.14)	0.023* (1.93)	0.029*** (2.88)
<i>WSDI</i> × <i>I</i> {latitude ≥ 40°}	−0.039** (−1.97)					
<i>WSDI</i> × <i>I</i> {latitude ≥ 45°}		−0.037* (−1.83)				
<i>WSDI</i> × <i>I</i> { <i>LSR1</i> ≤ 0.23}			−0.007 (−0.34)			
<i>WSDI</i> × <i>I</i> { <i>LSR2</i> ≤ 0.12}				−0.004 (−0.23)		
<i>Precip</i>					−0.000 (−1.14)	−0.000 (−0.81)
<i>Controls</i>	Yes	Yes	Yes	Yes	No	Yes
<i>Country/Year FE</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>Observations</i>	688	688	688	688	691	685
<i>Countries</i>	48	48	48	48	49	48

Additionally, the combined value of these coefficients remains greater than zero. This suggests that for countries located above the latitude cutoff, the increase in sovereign bond yield spreads due to sustained extreme heat is moderated, although the total effect remains positive. For example, in Column (2), the total effect of extreme heat is 0.004%, which translates to a rise in sovereign bond yield spreads by approximately 0.07% annually at the annual average level of 17.02 extreme heat days, significantly lower than the 0.48% observed in the baseline regression..

Columns (3) and (4) test the differential impact of sea level rise risk. The cutoff point for the dummy variable is set at the sample median. It can be seen that, whether *LSR1* or *LSR2* is used to measure exposure to the risks associated with rising sea levels, the coefficient for the interaction term is negative but not significant. Columns (5) and (6) show the estimation results when controlling for precipitation and the results indicate that whether only precipitation is controlled for, or precipitation is added to the existing control variables, the estimated coefficient for the core explanatory variable fluctuates by only 0.001 and remains significant. These results suggest that the effects of extreme heat on sovereign bond yield spreads is not driven by its correlation with sea level rise or drought.

In conclusion, while extreme heat shows a strong correlation with sea level rise and drought, its effect on sovereign bond yield spreads is not dominated by this correlation, which further demonstrates that extreme heat is a typical representative of climate disasters.

4.2.3. Subsample test

The climate indicators used in this study are aggregated to the national level from data collected at all weather stations within each country. In the case of large countries, such averaging may dilute the risks of extreme heat. To address this concern, the four largest countries by land area were removed from the sample: Russia, Canada, China, and Brazil (the U.S. is not included in the sample), and Column (1) of Table 8 shows the related results, which indicating that the estimated coefficients are very close to those from the baseline regression and remain significantly positive at the 95% level, affirming that the findings drawn from the baseline regression still hold.

Additionally, the 10-year sovereign bond data for these four countries were substituted with data from bonds of other maturities. After removing these four countries from the sample (Column (2) of Table 8), the estimated coefficient for the core explanatory variable changes very little and remains significant. Similarly, the short-term interest rate in

Table 8
Subsample test.

Variable	(1) Excluding the 4 Largest Countries by Land Area	(2) Excluding the Substituted 10- Year Sovereign Bond Sample	(3) Excluding OECD Forecasted Short-Term Interest Rates	(4) High-Income Group	(5) Non-High- Income Group	(6) High-Income + Upper- Middle-Income Group
<i>L.Spread</i>	0.587*** (15.48)	0.582*** (19.89)	0.578*** (18.67)	0.448*** (9.58)	0.719*** (8.53)	0.574*** (14.99)
<i>WSDI</i>	0.031** (2.16)	0.027** (2.09)	0.029** (2.36)	0.009 (0.65)	0.065** (2.39)	0.032*** (2.60)
<i>Controls</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>Country/Year FE</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>Observations</i>	627	675	675	461	218	591
<i>Countries</i>	44	44	47	27	25	39

the sample were taken from 13 OECD forecasts. To demonstrate that including these forecasts does not significantly affect the estimation results, we removed these observations in Column (3) of Table 8, and the results remain robustness.

A country's capacity to withstand natural disasters is related to its income level, with higher-income countries generally perceived as possessing greater disaster resilience. To investigate if the effect of extreme heat on sovereign bond yield spreads different among various income groups, we divide the sample into multiple subsamples based on the World Bank's income group classification from 1995 to 2018, which classifies countries into four categories: high-income, upper-middle-income, lower-middle-income, and low-income groups.¹¹ Column (4) of Table 8 reports the estimation results for the high-income group. The estimated coefficient on the Warm Spell Duration Index is 0.009 and statistically insignificant, indicating that, within our sample, extreme heat is not strongly associated with sovereign bond yield spreads for high-income economies. Column (5) presents the estimates for the non-high-income group. In this case, the coefficient on the Warm Spell Duration Index rises to 0.065 and is significant at the 5% level, suggesting a stronger association between extreme heat and sovereign spreads for non-high-income countries. Column (6) combines high-income and upper-middle-income economies; the coefficient on the Warm Spell Duration Index becomes significant once the upper-middle-income group is added to the high-income sample. Taken together, these patterns are consistent with the idea that, in our sample, lower-income and less resilient sovereigns tend to exhibit greater sensitivity of bond yield spreads to extreme heat. However, given the limited number of observations for low-income countries, these results should be viewed as suggestive and may not fully represent the broader universe of developing economies..

4.2.4. Changing the estimation method

The baseline regression results are derived from the LSDVC model. To validate the robustness of the findings, alternative methodologies are employed to estimate the equation, with results detailed in Table 9. It is crucial to emphasize that the estimation results presented in Columns (1) and (2) of Table 9 provide the upper and lower bounds for the estimated coefficient of the lagged spread, specifically [0.497, 0.632], and the validity of any other estimation method can be confirmed only if its estimated coefficient for the lagged spread falls within these bounds. Columns (3) and (4) in Table 9 represent the estimation results obtained through the difference GMM and system GMM approaches, respectively. Both methods pass the over-identification test and the second-order autocorrelation test, and the estimated coefficients for the lagged spread lie within the bounds, satisfying the estimation conditions. Models (5) and (6) incorporate credit ratings by using the LSDVC model with initial values based on the estimators of Blundell and Bond (1998)

and Anderson and Hsiao (1981), respectively, and the estimated coefficients for the lagged spread also lie within the bounds. Besides, the regression outcomes in Columns (3)–(6) indicate that the estimated coefficients for the Warm Spell Duration Index are all significantly positive, ranging from 0.025 to 0.029, which suggests that the empirical findings from the baseline regression are not unique to a specific estimation method. In column (7), we replace the one-period lagged control variables for deficit (*Deficit*) and debt (*Debt*) in the baseline regression with their contemporaneous values, and the coefficient and significance level of WSDI remain unchanged..

4.3. Channel test

The previous robustness analyses confirmed that extreme heat increases sovereign bond yield spreads, and validate the conclusions of the theoretical model. Next, this study further examines the channels that extreme heat affects sovereign bond yield spreads from the perspective of agriculture and social stability.

4.3.1. Agriculture channel

If the impact of extreme heat on sovereign bond yield spreads occurs through damage to agriculture, then the effect of extreme heat should be stronger for samples with higher agricultural dependence, leading to a greater increase in sovereign bond yield spreads. This study uses the proportion of agriculture in GDP (referred to as ATG) from the World Bank's open database to represent the agricultural dependence of the local economy, and use two methods to test the agriculture channel. First, the interaction term of agricultural proportion and the Warm Spell Duration Index is added to the baseline model. To better interpret the coefficient values, the agricultural proportion is standardized. The estimation equation is as follows:

$$spread_{it} = \beta_0 + \beta_1 spread_{it-1} + \beta_2 WSDI_{it} + \gamma_1 WSDI_{it} * ATG_{-sit} + \eta_1 ATG_{-sit} + \tau z_{it} + A_i + B_i + \varepsilon_{it}$$

where ATG_{-sit} represents the standardized agricultural proportion for country i in year t , and γ_1 is the coefficient of interest. Column (1) in Table 10 presents the relevant estimation results, showing a positive yet statistically insignificant coefficient for the interaction term, which does not necessarily mean that the agricultural proportion has no impact, rather, it suggests that the effect may be non-linear. Consequently, this study uses interaction terms with grouped dummy variables to further validate these findings. The results are presented in Columns (2) and (3), in which the threshold at an average agricultural proportion of 6% and 5.5%, respectively. The findings indicate that the coefficient for the interaction term is significantly positive, while the coefficient for the core explanatory variable becomes insignificant, indicating that the influence of extreme heat on sovereign bond yield spreads mainly comes from countries with higher average agricultural proportion.

Since the correlation coefficient between the agricultural share of value added and per capita GDP (GDP_{pc}) is -0.574 (Appendix Table A2), and Table 8 indicates that the estimated effect of extreme heat on

¹¹ The group classification is primarily based on per capita income. For detailed thresholds, please refer to the official World Bank documentation.

Table 9

Changing the estimation method.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Variable	OLS	FE	DGMM	SGMM	LSDVC_ab	LSDVC_ah	LSDVC
<i>L.Spread</i>	0.632*** (5.98)	0.497*** (6.04)	0.552*** (5.11)	0.612*** (7.42)	0.568*** (15.49)	0.572*** (15.31)	0.580*** (15.54)
<i>WSDI</i>	0.027** (2.34)	0.027* (1.70)	0.025** (2.06)	0.029** (2.32)	0.027*** (2.66)	0.028** (2.57)	0.028*** (2.67)
<i>Controls</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Country/Year FE</i>	No	Yes	Yes	Yes	Yes	Yes	Yes
<i>Observations</i>	688	688	631	688	688	688	688
<i>Countries</i>	48	48	46	48	48	48	48
<i>Sargan Test p-value</i>			0.910	0.897			
<i>AR(2) Test p-value</i>			0.330	0.205			
<i>Wald Test p-value</i>			0.000	0.000			

Note: Columns (1) and (2) represent OLS regression and fixed-effects regression, respectively, with robust standard errors applied. Columns (3) and (4) represent two-step robust difference GMM and system GMM models, with the instruments being the second and higher-order lags of the spread, as well as the other variables themselves. Columns (5) and (6) represent LSDVC models with initial values based on the estimators of [Blundell & Bond \(1998\)](#) and [Anderson & Hsiao \(1981\)](#), with 50 iterations of bootstrapping and bias precision of $o(n^{-1}T^{-2})$. *t*-values are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 10

Agriculture channel.

	(1)	(2)	(3)	(4)	(5)
Variable	Full Sample	Full Sample	Full Sample	Full Sample	FE
<i>L.Spread</i>	0.577*** (19.30)	0.580*** (16.15)	0.584*** (16.42)	0.579*** (15.39)	
<i>WSDI</i>	0.028** (2.22)	0.016 (1.40)	0.011 (0.94)	0.012 (0.83)	−0.001** (−2.02)
<i>WSDI</i> × <i>ATG_s</i>	−0.002 (−0.19)				
<i>WSDI</i> × <i>I</i> { <i>mATG</i> ≥ 6}		0.043* (1.73)			
<i>WSDI</i> × <i>I</i> { <i>ATG</i> ≥ 5.5}			0.052** (2.20)	0.092*** (3.18)	
<i>Controls</i>	Yes	Yes	Yes	Yes	
<i>Country/Year FE</i>	Yes	Yes	Yes	Yes	
<i>Observations</i>	686	688	688	591	687
<i>Countries</i>	48	48	48	39	48

Note: Control variables include macroeconomic variables, maximum temperature, and the variables involved in the interaction terms.

sovereign bond yield spreads is stronger for the non-high-income subsample in our data, one may wonder whether this pattern simply reflects the higher agricultural dependence of countries that we classify as “developing.” To explore this issue, we re-estimate the model on a restricted sample consisting only of high-income and upper-middle-income economies, where income levels are relatively higher but agricultural dependence still varies across countries. The results are reported in Column (4) of [Table 10](#) and remain qualitatively similar. Across the four specifications, countries with greater agricultural dependence tend to exhibit a larger estimated sensitivity of sovereign bond yield spreads to extreme heat. These findings suggest that agricultural structure is an important dimension of vulnerability in our sample, although, given the limited coverage of low-income economies, the evidence should be interpreted as indicative rather than fully representative of all developing countries.

To demonstrate that extreme heat affects sovereign bond yield spreads through the agricultural channel, in column (5) we use the natural logarithm of agricultural value added for each country, obtained from the World Bank database, as the dependent variable and estimate a fixed-effects model. The empirical results show that extreme heat (*WSDI*) significantly reduces agricultural output at the 5% significance level.

4.3.2. Social stability channel

If the influence of extreme heat on sovereign bond yield spreads is attributable to its effects on social stability, then this impact should be more pronounced in samples where social stability is already compromised. This research utilizes crime rate data (*Crime*) sourced from the World Bank database, which records the number of crimes per 100,000 people, as a measure of social stability. Similar to the methodology used to test the agriculture channel, the sample first uses the interaction term between the standardized annual crime rate and the Warm Spell Duration Index for estimation, with the estimation equation similar to Eq. (11). The standardized crime rate (*Crime_s_{it}*) is also used.

Unlike the validation of the agriculture channel, the hypothesis is that extreme heat reduces social stability, i.e., increases the crime rate. When the crime rate is added to the model, Column (1) of [Table 11](#) clearly shows the estimated coefficient for the Warm Spell Duration Index decreases by 0.002, with decreasing significance level. Column (2) presents the results for the interaction term, with an estimated coefficient of 0.017 that is significant at the 99% level, suggesting that for each one standard deviation increase in the crime rate, the effect of extreme heat on sovereign bond yield spreads increases by 1.7%.

What's more, it is important to highlight that the aforementioned estimation results may be influenced by the positive correlation between extreme heat and crime rates. To reduce this concern, we further use the average crime rate during the sample period to represent average social stability and applies interaction terms with grouped dummy variables for verification. Here, the threshold for the high crime rate group is set at

Table 11

Social stability channel.

	(1)	(2)	(3)	(4)	(5)
Variable	Full Sample	Full Sample	Full Sample	Full Sample	FE
<i>L.Spread</i>	0.577*** (14.42)	0.578*** (14.80)	0.580*** (16.15)	0.574*** (15.13)	
<i>WSDI</i>	0.027** (2.18)	0.023* (1.82)	0.019* (1.81)	0.020 (1.60)	0.003* (1.85)
<i>Crime_s</i>	0.287 (0.61)	−0.101 (−0.21)			
<i>WSDI</i> × <i>Crime_s</i>		0.017*** (2.82)			
<i>WSDI</i> × <i>I</i> { <i>mCrime</i> ≥ 5}			0.039* (1.70)	0.053** (1.97)	
<i>Controls</i>	Yes	Yes	Yes	Yes	Yes
<i>Country/Year FE</i>	Yes	Yes	Yes	Yes	Yes
<i>Observations</i>	658	658	688	591	687
<i>Countries</i>	45	45	48	39	48

5 crimes per 100,000 people, and the estimation results are presented in Column (3) of Table 11, which indicate that both the estimated coefficients for the Warm Spell Duration Index and the interaction term are significantly positive, suggesting that in countries with higher average crime rates, extreme heat leads to a more significant rise in sovereign bond yield spreads.

Similar to the analysis of agricultural dependence, Appendix Table A2 shows that the correlation coefficient between crime rate and per capita GDP is -0.365 , suggesting that developing countries generally exhibit higher average crime rates, which may dominate the model's estimation results. Similarly, we re-estimate the model with a subsample of high-income and upper-middle-income countries, finding that the estimated coefficient for the interaction term in Column (4) of Table 11 continues to be significantly positive, resulting in comparable conclusions. To verify the social stability channel, in column (5) we use the number of crimes per 100,000 people in each country, obtained from the World Bank database, as the dependent variable and estimate a fixed-effects model. The empirical results show that extreme heat (WSDI) significantly increases the crime rate at the 10% significance level, indicating that extreme heat undermines social stability.

In conclusion, we have validated the key assumption of the theoretical model: extreme heat elevates the probability of climate disasters by damaging agriculture and social stability, thereby increasing sovereign bond yield spreads.

5. Strategies for mitigating the effects of extreme heat

5.1. Climate response capacity

To examine whether the climate response capacity of different countries can reduce the negative impacts of extreme heat, this study uses the publicly available data from the Notre Dame Global Adaptation Index (ND-GAIN) to derive the Climate Vulnerability and Climate Readiness indices for over 180 countries.¹² Climate vulnerability refers to the extent to which human society is susceptible to the negative effects of climate risks, which is determined by considering 36 indicators across six sectors: food, water resources, health, ecosystem services, human living conditions, and infrastructure. Climate readiness refers to a nation's capacity to attract and utilize investments for adaptive actions and is comprised of 9 indicators across three areas: economic, governance, and social.¹³ Table 12 shows characteristics of the related data.

Fig. 3 illustrates the trends in climate vulnerability and climate readiness for five selected countries: China, Germany, India, Canada, and Australia. As depicted in Fig. 3, the trends in climate vulnerability

Table 12
Climate response capacity-related indicators.

Variable	Indicator Name	Obs	Mean	Sd	Min	Max
<i>vulnerability</i>	Climate Vulnerability	757	0.360	0.060	0.250	0.550
<i>readiness</i>	Climate Readiness	757	0.570	0.150	0.260	0.810
<i>food</i>	Food	757	0.320	0.110	0.170	0.600
<i>wate</i>	Water Resources	757	0.340	0.090	0.170	0.690
<i>heal</i>	Health	757	0.330	0.100	0.180	0.710
<i>ecos</i>	Ecosystem Services	757	0.390	0.090	0.230	0.590
<i>habi</i>	Human Living Conditions	757	0.450	0.070	0.330	0.650
<i>infr</i>	Infrastructure	757	0.330	0.100	0.080	0.570
<i>business</i>	Economy	757	0.580	0.160	0.180	0.850
<i>gove</i>	Governance	757	0.650	0.170	0.250	0.900
<i>soci</i>	Society	757	0.480	0.160	0.150	0.810

¹² The composition of these indices is presented in website: <https://gain.nd.edu/our-work/country-index/>.

¹³ Sources: <https://gain.nd.edu/our-work/country-index/>.

not only vary in direction over time but also in their initial levels. Both China and India, as developing countries, exhibit a downward trend in climate vulnerability from a high point during the sample period, while their climate readiness demonstrates an upward trend from a lower baseline. In contrast, Germany, Canada, and Australia, three developed countries, have consistently maintained lower levels of climate vulnerability and higher levels of climate readiness.

Furthermore, we do some empirical tests following Eqs. (10) and (11). Table 13 show the relevant results, with the first two columns show the results for climate vulnerability, while the last two columns show the results for climate readiness. Column (1) of Table 13 initially assesses the heterogeneous effect of standardized climate vulnerability (*Vul_s*), where the estimated coefficient for the interaction term is positive but not significant. Consequently, in Column (2) of Table 13, we then employ an interaction term with a dummy variable, setting the cutoff at 0.36 (the sample mean). In this case, the estimated coefficient for the interaction term is significantly negative, indicating that climate vulnerability itself does not affect sovereign bond yield spreads, but only when it interacts with extreme heat does it have an impact.

Next, this study continues to test the effect of climate readiness. Column (3) of Table 13 presents the estimation findings of the interaction term between the standardized climate readiness (*Read_s*) and the Warm Spell Duration Index. Although the coefficient estimated for climate readiness is not significant, the interaction term has a coefficient of -0.022 , which is significant at the 95% level. This suggests that a one standard deviation increase in climate readiness leads to an overall effect of 0.003% on sovereign bond yield spreads due to extreme heat. To further test whether countries with higher climate readiness can mitigate the impact of extreme heat on sovereign bond yield spreads, this study sets two thresholds to divide the sample into three groups: 0.57 and 0.72, with 0.57 close to the sample mean and 0.72 near the sample maximum of 0.81. Two dummy variables, $I\{0.72 > readiness \geq 0.57\}$ and $I\{readiness \geq 0.72\}$, represent samples with climate readiness between [0.57, 0.72) and [0.72, 0.81], respectively. Interaction terms with the Warm Spell Duration Index are then added to the model, and the Column (4) in Table 13 shows the corresponding estimation results, which indicates that when extreme heat occurs, sovereign bond yield spreads increase by 0.002% for samples with climate readiness between [0.57, 0.72), while for samples with climate readiness above 0.72, sovereign bond yield spreads decrease by 0.005%. This suggests that when climate readiness is sufficiently high, extreme heat actually leads to a reduction in sovereign bond yield spreads.

In summary, climate response capacity itself does not have a direct influence on sovereign bond yield spreads, but it modifies the impact of extreme heat. Efforts by countries to enhance their climate response capacity are effective, as they can significantly mitigate the increase in sovereign bond yield spreads attributable extreme heat. This finding for the high-readiness countries can be rationalized by considering the portfolio-reallocation behavior of global investors. During episodes of heightened climate stress, investors may engage in a “flight-to-safety” within the sovereign bond market, reallocating capital from more vulnerable countries to those perceived as safer because of their stronger climate preparedness and institutional quality. In line with the broader literature on flight-to-quality in sovereign and government bond markets (e.g., Beber, Brandt, & Kavajecz, 2009; Caballero & Krishnamurthy, 2008), such a reallocation would compress yields—and thus spreads—in high-readiness countries exactly when climate shocks occur. In addition, high climate readiness may signal more credible adaptation policy and greater fiscal capacity to absorb climate shocks, so that extreme heat events are interpreted by markets as less damaging to the long-run default risk of these sovereigns. We therefore view the negative interaction effect observed for the high-readiness group as being consistent with a combination of “flight-to-safety” dynamics and a lower climate-disaster risk premium in countries with very strong climate response capacity..

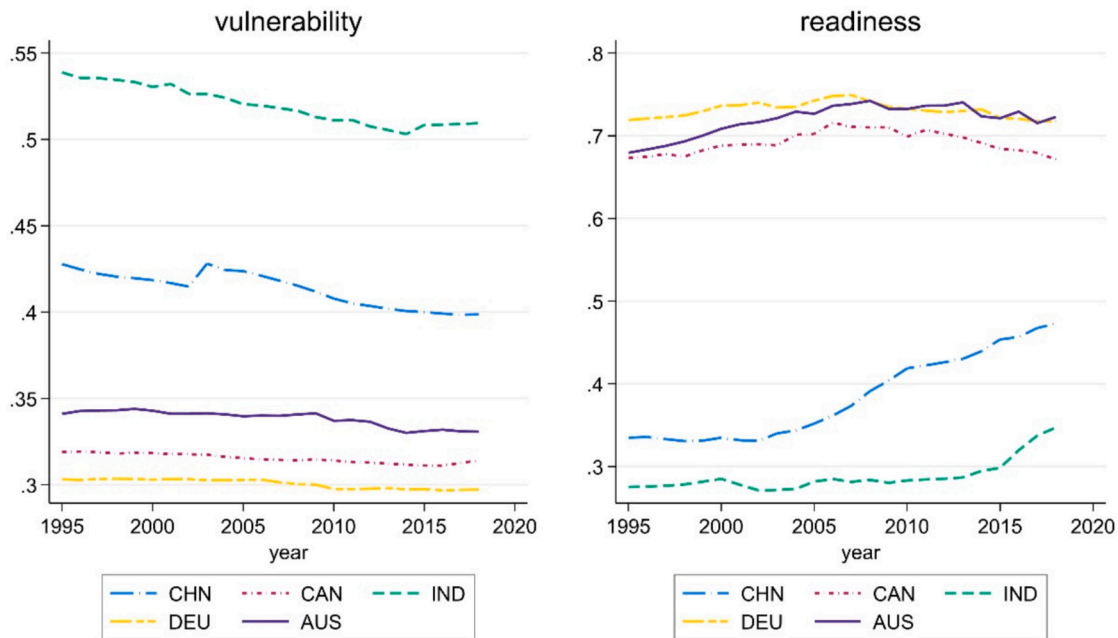


Fig. 3. Climate vulnerability and climate readiness.

Table 13
Effectiveness of climate response capacity.

Variable	(1) Climate Vulnerability	(2) Climate Vulnerability	(3) Climate Readiness	(4) Climate Readiness
<i>L.Spread</i>	0.578*** (16.10)	0.559*** (15.55)	0.578*** (15.98)	0.581*** (16.38)
<i>WSDI</i>	0.028*** (2.76)	0.068*** (3.60)	0.025** (2.46)	0.047*** (3.58)
<i>Vul_s</i>	-1.975 (-1.26)			
<i>WSDI × Vul_s</i>	0.017 (1.26)			
<i>WSDI × I{Vulnerability ≤ 0.36}</i>		-0.058*** (-2.63)		
<i>Read_s</i>			0.963 (1.04)	
<i>WSDI × Read_s</i>			-0.022* (-1.73)	
<i>WSDI × I{Readiness ≥ 0.72}</i>				-0.052* (-1.67)
<i>WSDI × I{0.72 > Readiness ≥ 0.57}</i>				-0.045** (-2.07)
<i>Controls</i>	Yes	Yes	Yes	Yes
<i>Country/Year FE</i>	Yes	Yes	Yes	Yes
<i>Observations</i>	687	687	687	687
<i>Countries</i>	48	48	48	48

5.2. Identifying key indicators for mitigating extreme heat impacts

The previous analysis showed that climate response capacity can effectively reduce the impact of extreme heat. However, the indicators represented by climate vulnerability and climate readiness measure overall climate response capacity, addressing all physical risks rather than focusing solely on the impact of extreme heat. Since the climate vulnerability score is derived from the arithmetic average of six sectors, and the climate readiness score is based on the arithmetic average of three sectors, this study will analyze the role of specific sectors to identify which ones play a key role in mitigating extreme heat. For example, when examining the role of the food sector, the analysis will focus exclusively on the vulnerability score for the food sector. It is important to note that for simplicity and directness, only interaction

terms with dummy variables will be used for testing, and the estimation equation will be similar to Eq. (10). The choice of cutoff points will first consider the sample mean. If the sign of the interaction term's estimated coefficient aligns with expectations but is not significant, the sample median will be used. If the sign remains consistent with expectations but is still not significant, other reference points will be considered.

The descriptive statistics for climate vulnerability across the six sectors are presented in Table 14, and the average value for all sectors in the sample is below 0.5, indicating a relatively low global level. Columns (1) to (6) of Table 14 represent the impact of the vulnerability of the food, water resources, health, ecosystem, living conditions, and infrastructure sectors on the harm caused by extreme heat. The results show that only the interaction terms for the food, health, and living conditions sectors have significant estimated coefficients, while the interaction terms for the other sectors are not significant (although Table 14 reports only one cutoff point, this study tested multiple cutoff points for the non-significant sectors, and the results remained insignificant). This suggests that, in the context of extreme heat, the vulnerabilities in the food, health, and living conditions sectors play a key role.

Specifically, the climate vulnerability of the food sector is closely linked to agriculture, such as agricultural capacity, which includes irrigation capability, fertilizer usage, pesticide usage, and the level of tractor utilization. These four aspects reflect the stage of agricultural technology, which is an important indicator of agricultural adaptability. The climate vulnerability of the health sector is closely tied to medical services, such as access to adequate sanitation facilities, a commonly used indicator for infectious disease control. Good sanitation facilities are especially crucial for preparedness against the natural disasters intensified by climate change. The living conditions sector emphasizes conditions related to natural disasters, such as the dependency ratio for vulnerable populations. Populations under the age of 14 and over the age of 65 are the most vulnerable groups, and extreme heat will have the most severe impact on these groups.

For the sub-sectors of climate readiness, the descriptive statistics for the readiness of the three sub-sectors are also presented in Table 12. In contrast to climate vulnerability, the climate readiness of all sectors in the sample is above 0.5, indicating a relatively high level. According to the official ND-GAIN documentation, both the business environment and management readiness reflect a country's ability to attract adaptive investments, while social readiness reflects the society's ability to fully

Table 14
Vulnerability of specific sectors.

Variable	(1) Food	(2) Water Resources	(3) Health	(4) Ecosystem	(5) Living Conditions	(6) Infrastructure
<i>Spread</i>	0.566*** (15.74)	0.578*** (15.66)	0.580*** (15.91)	0.579*** (16.15)	0.574*** (16.17)	0.579*** (16.08)
<i>WSDI</i>	0.051*** (3.77)	0.027** (2.22)	0.043*** (3.58)	0.029* (1.81)	0.046*** (3.75)	0.033*** (2.62)
<i>WSDI</i> × <i>I</i> { <i>food</i> ≤ 0.32}	−0.049*** (−2.67)					
<i>WSDI</i> × <i>I</i> { <i>wate</i> ≤ 0.30}		0.001 (0.07)				
<i>WSDI</i> × <i>I</i> { <i>heal</i> ≤ 0.28}			−0.047** (−2.14)			
<i>WSDI</i> × <i>I</i> { <i>ecos</i> ≤ 0.40}				−0.002 (−0.11)		
<i>WSDI</i> × <i>I</i> { <i>habi</i> ≤ 0.43}					−0.036* (−1.75)	
<i>WSDI</i> × <i>I</i> { <i>infr</i> ≤ 0.31}						−0.015 (−0.88)
<i>Controls</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>Country</i> / <i>Year FE</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>Observations</i>	687	687	687	687	687	687
<i>Countries</i>	48	48	48	48	48	48

utilize such investments. The business environment refers to the degree, time, and cost involved in conducting business activities, and includes 10 themes, such as access to electricity, access to credit, investor protection, and contract enforcement, and is frequently used in studies to assess the investment environment. Management readiness emphasizes the government's administrative capacity, such as the quality of governance, which represents the government's capability to formulate and implement effective adaptive policies and regulations. Social readiness highlights the society's ability to fully utilize investments, such as the level of education, which is considered a crucial strategy for building adaptive capacity and formulating localized solutions.

Table 15 displays the estimation results for the interaction terms. Columns (1) to (3) correspond to business environment, management readiness, and social readiness, with the cutoff points chosen as the mean, median, and mean, respectively. The results show that the interaction terms between WSDI and business environment (column (1)) and between WSDI and social readiness (column (3)) are significantly negative, whereas the interaction with management readiness (column (2)) is negative but not statistically significant. These findings suggest that better business and social readiness can effectively dampen the impact of extreme heat on sovereign bond yield spreads, while the

moderating role of management readiness is less robust.

In conclusion, improvements in the food, health, and living conditions sectors significantly contribute to alleviating the adverse effects of extreme heat. Regarding climate readiness, the evidence indicates that, among the three dimensions considered, business environment and social readiness play particularly important roles in mitigating the impact of extreme heat on sovereign bond yield spreads, whereas the independent moderating effect of management readiness is relatively weak. Adaptive investments serve as the foundation for enhancing climate response capacity, and a robust business environment and effective government administrative capacity are essential conditions for attracting such investments. Once these investments have been secured, their optimal utilization becomes imperative; thus, the significance of societal education and innovation capacity must not be overlooked..

6. Conclusions

Climate change is one of the most pressing issues in society today, and even developed economies cannot overlook the risks associated with climate-related disasters. By incorporating extreme heat into an asset pricing model and using panel climate data from multiple countries between 1995 and 2018 to do empirical tests, this study explores the influence of climate disaster risk on sovereign bond yield spreads, focusing specifically on extreme heat. The key findings are as follows:

First, extreme heat raises sovereign bond yield spreads, particularly in developing countries. The empirical results remain robust across various estimation methods. Although extreme heat is correlated with sea level rise and drought, the estimation results are not driven by these correlations. For the Warm Spell Duration Index with a sample mean of 16.78 days, sovereign bond yield spreads increase by approximately 0.47% annually.

Second, the primary channels through which extreme heat affects sovereign bond yield spreads are agriculture and social stability. Countries with higher agricultural dependence or crime rates face greater risk compensation in their sovereign bond yield spreads due to extreme heat. While both agricultural dependence and crime rates are correlated with per capita GDP, this correlation does not drive the conclusions, and the results hold even in upper-middle-income and higher-income countries.

Third, with climate vulnerability and climate readiness as proxy indicators for climate response capacity, this study finds that climate response capacity can effectively reduce the adverse impacts of extreme heat. Regarding the alleviating harm caused by extreme temperatures,

Table 15
Readiness of specific sectors.

Variable	(1) Business Environment	(2) Management Readiness	(3) Social Readiness
<i>Spread</i>	0.582*** (16.31)	0.577*** (15.96)	0.584*** (16.96)
<i>WSDI</i>	0.049*** (3.82)	0.041*** (3.03)	0.043*** (3.41)
<i>WSDI</i> × <i>I</i> { <i>busi</i> ≥ 0.58}	−0.050** (−2.34)		
<i>WSDI</i> × <i>I</i> { <i>gove</i> ≥ 0.67}		−0.036 (−1.58)	
<i>WSDI</i> × <i>I</i> { <i>soci</i> ≥ 0.55}			−0.043** (−2.20)
<i>Controls</i>	Yes	Yes	Yes
<i>Country</i> / <i>Year FE</i>	Yes	Yes	Yes
<i>Observations</i>	687	687	687
<i>Countries</i>	48	48	48

vulnerabilities within food security, healthcare systems, and living conditions—as well as readiness within business environments and societal engagement—exert significant effects.

Drawing from the research findings mentioned above, this study suggests the following policy recommendations:

First, sovereign risk management should explicitly incorporate extreme-heat risk. Debt management offices and central banks can use indicators such as the Warm Spell Duration Index (WSDI) to run climate stress tests within debt sustainability analysis and sovereign asset-liability management frameworks. Scenario analysis with more frequent and intense heatwaves can then inform target debt levels, maturity structure and currency composition, as well as the size of precautionary fiscal buffers, rather than treating climate shocks as exogenous and rare residual events.

Second, governments should employ concrete sovereign risk-management tools that create fiscal space when extreme-heat shocks materialize. This includes issuing catastrophe bonds or parametric insurance contracts whose payouts are linked to extreme-heat indices, introducing climate-contingent clauses in sovereign bonds (e.g. temporary standstills or coupon step-downs triggered by officially declared heat-related disasters), and arranging contingent credit lines with multilateral institutions that disburse automatically after severe heat

events. These instruments can smooth debt-service profiles, reduce the need for abrupt fiscal tightening in the aftermath of shocks, and thereby limit the increase in sovereign spreads documented in our empirical analysis.

Third, fiscal policy should prioritize adaptation investments in the specific sectors that our results identify as mitigating the impact of extreme heat on sovereign spreads—namely food and agriculture, health systems, living conditions, the business environment and social readiness. In practical terms, this implies protecting and scaling up spending on climate-resilient agriculture, heat-related public health capacity, housing and urban infrastructure that reduce heat exposure, as well as institutional quality and social cohesion. Embedding these priorities into medium-term expenditure frameworks and climate-budget-tagging systems can simultaneously enhance resilience and lower sovereign borrowing costs in countries most exposed to extreme heat.

Funding information

National Social Science Fund Youth Project Grant numbers: 24CJL065 China Postdoctoral Science Foundation Grant numbers: 2025M783682 Guangdong University of Foreign Studies Grant numbers: 25TS04

Appendix A

Table A1

Distribution of country or region samples.

Country	Obs	Proportion	Cumulative Proportion	Country	Obs	Proportion	Cumulative Proportion
Australia	24	3.17	3.17	China	16	2.11	73.45
Belgium	24	3.17	6.34	Czech	16	2.11	75.56
Canada	24	3.17	9.51	Israel	16	2.11	77.68
France	24	3.17	12.68	Russian Fed	15	1.98	79.66
Germany	24	3.17	15.85	Colombia	14	1.85	81.51
Italy	24	3.17	19.02	Mexico	14	1.85	83.36
Japan	24	3.17	22.19	Iceland	12	1.59	84.94
Netherlands	24	3.17	25.36	Brazil	11	1.45	86.39
Portugal	24	3.17	28.53	Malaysia	11	1.45	87.85
Spain	24	3.17	31.7	Pakistan	10	1.32	89.17
United Kingdom	24	3.17	34.87	Vietnam	10	1.32	90.49
Austria	23	3.04	37.91	Chile	9	1.19	91.68
New Zealand	23	3.04	40.95	Croatia	9	1.19	92.87
Switzerland	23	3.04	43.99	Turkey	8	1.06	93.92
Norway	22	2.91	46.9	Egypt	7	0.92	94.85
South Africa	21	2.77	49.67	Indonesia	7	0.92	95.77
Greece	20	2.64	52.31	Slovakia	7	0.92	96.7
India	20	2.64	54.95	Morocco	6	0.79	97.49
Poland	19	2.51	57.46	Namibia	4	0.53	98.02
Hungary	18	2.38	59.84	Ukraine	4	0.53	98.55
Ireland	18	2.38	62.22	Argentina	3	0.4	98.94
Korea (South)	18	2.38	64.6	Bangladesh	3	0.4	99.34
Philippines	18	2.38	66.97	Botswana	3	0.4	99.74
Thailand	17	2.25	69.22	Jordan	2	0.26	100
Bulgaria	16	2.11	71.33				

Table A2

Correlation table of other variables.

Variable	GDP _{pc}	ATG	Crime	Vulnerability	Readiness
GDP _{pc}	1.000				
ATG	−0.574	1.000			
Crime	−0.365	0.115	1.000		
Vulnerability	−0.659	0.828	0.335	1.000	
Readiness	0.789	−0.704	−0.467	−0.774	1.000

Appendix B

B.1. Solving Eq. (4)

First, we need to pre-calculate the required marginal density function and joint density function. The probability density of $B_{b,t+1}$ is

$$B_{b,t+1} = \begin{cases} 1 & \text{nodefault} & P\{B_{b,t+1} = 1\} = e^{-\lambda_t} \left(\frac{1}{e} - 1\right) \\ 1 - d & \text{default} & P\{B_{b,t+1} = 1 - d\} = 1 - e^{-\lambda_t} \left(\frac{1}{e} - 1\right) \end{cases} \quad (\text{A1})$$

Where,

$$\begin{aligned} P\{B_{b,t+1} = 1\} &= \sum_k P\{B_{b,t+1} = 1, N_{t+1} = k\} \\ &= \sum_k P\{B_{b,t+1} = 1 | N_{t+1} = k\} \cdot P\{N_{t+1} = k\} \\ &= \sum_k e^{-k\lambda_t} \frac{\lambda_t^k}{k!} e^{-\lambda_t} = e^{-\lambda_t} \left(\frac{1}{e} - 1\right) \end{aligned} \quad (\text{A2})$$

The joint probability density of $(B_{b,t+1}, N_{t+1})$ at this point is:

$$P\{B_{b,t+1} = 1, N_{t+1} = k\} = P\{B_{b,t+1} = 1 | N_{t+1} = k\} \cdot P\{N_{t+1} = k\} = e^{-k\lambda_t} \frac{\lambda_t^k}{k!} \quad (\text{A3})$$

$$\begin{aligned} P\{B_{b,t+1} = 1 - d, N_{t+1} = k\} &= P\{B_{b,t+1} = 1 - d | N_{t+1} = k\} \cdot P\{N_{t+1} = k\} \\ &= (1 - e^{-k\lambda_t}) \frac{\lambda_t^k}{k!} e^{-\lambda_t} \end{aligned} \quad (\text{A4})$$

Now, solving from the Euler equation:

$$\begin{aligned} P_{b,t} &= E_t \left[e^{-\delta} \left(\frac{C_{t+1}}{C_t} \right)^{-\theta} B_{b,t+1} \right] \\ &= E_t \left[e^{-\delta - \theta(\ln C_{t+1} - \ln C_t) + \ln B_{b,t+1}} \right] \\ &= E_t \left[e^{-\delta - \theta(\gamma + u_{t+1} + N_{t+1} \log(1-d)) + \ln B_{b,t+1}} \right] \\ &= e^{-\delta - \theta\gamma + \frac{1}{2}\theta^2\sigma_u^2} \cdot \left[\sum_{k=0}^{\infty} e^{-\theta k \log(1-d) + \ln 1} \cdot e^{-k\lambda_t} \frac{\lambda_t^k}{k!} e^{-\lambda_t} + \sum_{k=0}^{\infty} e^{-\theta k \log(1-d) + \ln(1-d)} \cdot (1 - e^{-k\lambda_t}) \frac{\lambda_t^k}{k!} e^{-\lambda_t} \right] \\ &= e^{-\delta - \theta\gamma + \frac{1}{2}\theta^2\sigma_u^2} \left[(1-d) e^{\lambda_t[(1-d)^{-\theta} - 1]} + d e^{\lambda_t \left[\frac{1}{e}(1-d)^{-\theta} - 1 \right]} \right] \end{aligned} \quad (\text{A5})$$

Taking the logarithm of this equation, we can obtain Eq. (4). During the calculation, the exponential expectation of the Poisson distribution $E_t[e^{\lambda_t N_{t+1}}]$ is used:

$$E_t[e^{\lambda_t N_{t+1}}] = \sum_{k=0}^{\infty} e^{\lambda_t k} \frac{\lambda_t^k}{k!} e^{-\lambda_t} = \sum_{k=0}^{\infty} \frac{(e^{\lambda_t} \lambda_t)^k}{k!} e^{-\lambda_t} (e^{\lambda_t} - e^{\lambda_t} + 1) = e^{\lambda_t(e^{\lambda_t} - 1)} \quad (\text{A6})$$

A2.2. Solving Eq. (5):

Based on Eq. (4), we only need to solve for $\ln E_t(B_{b,t+1})$:

$$\begin{aligned} \ln E_t(B_{b,t+1}) &= \ln(1) \sum_k P\{B_{b,t+1} = 1, N_{t+1} = k\} + \ln(1-d) \sum_k P\{B_{b,t+1} = 1-d, N_{t+1} = k\} \\ &= \ln(1-d) \sum_k (1 - e^{-k\lambda_t}) \frac{\lambda_t^k}{k!} e^{-\lambda_t} \\ &= \ln(1-d) \cdot \left(1 - e^{\lambda_t \left(\frac{1}{e} - 1 \right)} \right) \end{aligned} \quad (\text{A7})$$

Substitute this result and Eq. (5) into the polynomial of $\ln E_t(B_{b,t+1})$ and finally use the Taylor expansion to obtain the approximate result:

$$\begin{aligned}
\ln E_t(R_{b,t+1}) &= \ln E_t\left(\frac{B_{b,t+1}}{P_{b,t}}\right) = \ln E_t(B_{b,t+1}) - \ln(P_{b,t}) \\
&= \ln(1-d) \cdot \left(1 - e^{\lambda_t \left(\frac{1}{e}-1\right)}\right) - \ln \delta + \theta \gamma - \frac{1}{2} \theta^2 \sigma_u^2 - \ln \left[(1-d) e^{\lambda_t [(1-d)^{-\theta}-1]} + d e^{\lambda_t \left[\frac{1}{e}(1-d)^{-\theta}-1\right]} \right] \\
&\approx d \lambda_t \frac{e-1}{e} \left[\frac{1}{(1-d)^{\theta}} - 1 \right] - \ln \delta + \theta \gamma - \frac{\theta^2}{2} \sigma_u^2 - (l_0 + l_1 WSDI_t) [(1-d)^{-\theta} - 1]
\end{aligned} \tag{A8}$$

When rare disasters do not lead to sovereign bond defaults, the bond is a risk-free asset ($B_{b,t+1} = 1$), and its Euler equation is:

$$P_{b,t} = E_t \left[e^{-\delta} \left(\frac{C_{t+1}}{C_t} \right)^{-\theta} \right] \tag{A9}$$

Thus, the expected return on sovereign bonds $\ln E_t(R_{b,t+1}^0)$ can be calculated at this point:

$$\begin{aligned}
\ln E_t(R_{b,t+1}^0) &= \ln E_t\left(\frac{1}{P_{b,t}}\right) = -\ln(P_{b,t}) \\
&= r_f - (l_0 + l_1 WSDI_t) [(1-d)^{-\theta} - 1]
\end{aligned} \tag{A10}$$

Where $r_f = -\ln \delta + \theta \gamma - \frac{\theta^2}{2} \sigma_u^2$. It can be observed that extreme heat leads to a decrease in sovereign bond yields. This is because the future payments of sovereign bonds are fixed, allowing them to hedge against the risk of rare disasters caused by extreme heat, making investors more inclined to hold sovereign bonds.

Data availability

Data will be made available on request.

References

- Anderson, T. W., & Hsiao, C. (1981). Estimation of dynamic models with error components. *Journal of the American statistical Association*, 76(375), 598–606.
- Andrade, S. C., Ekponon, A., & Jeanneret, A. (2023). Sovereign risk premia and global macroeconomic conditions. *Journal of Financial Economics*, 147(1), 172–197.
- Baldauf, M., Garlappi, L., & Yannelis, C. (2020). Does climate change affect real estate prices? Only if you believe in it. *The Review of Financial Studies*, 33(3), 1256–1295.
- Bansal, R., Kiku, D., & Ochoa, M. (2019). *Climate change risk. Working paper*.
- Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *The Quarterly Journal of Economics*, 121(3), 823–866.
- Barro, R. J. (2015). Environmental protection, rare disasters and discount rates. *Economica*, 82(325), 1–23.
- Bauer, M. D., Huber, D., Rudebusch, G. D., & Wilms, O. (2022). Where is the carbon premium? Global performance of green and brown stocks. *Journal of Climate Finance*, 1, Article 100006.
- Beber, A., Brandt, M. W., & Kavajecz, K. A. (2009). Flight-to-quality or flight-to-liquidity? Evidence from the euro-area bond market. *The Review of Financial Studies*, 22(3), 925–957.
- Blundell, R., & Bond, S. (1998). Initial conditions and moment restrictions in dynamic panel data models[J]. *Journal of econometrics*, 87(1), 115–143.
- Bolton, P., & Kacperczyk, M. (2021). Do investors care about carbon risk? *Journal of Financial Economics*, 142(2), 517–549.
- Bonato, M., Cepni, O., Gupta, R., & Pierdzioch, C. (2023). Climate risks and state-level stock market realized volatility. *Journal of Financial Markets*, 66, Article 100854.
- Burke, M., & Emerick, K. (2016). Adaptation to climate change: Evidence from US agriculture. *American Economic Journal: Economic Policy*, 8(3), 106–140.
- Caballero, R. J., & Krishnamurthy, A. (2008). Collective risk management in a flight to quality episode. *The Journal of Finance*, 63(5), 2195–2230.
- Capelle-Blancard, G., Crifo, P., Diaye, M. A., Oueghlissi, R., & Scholtens, B. (2019). Sovereign bond yield spreads and sustainability: An empirical analysis of OECD countries. *Journal of Banking & Finance*, 98, 156–169.
- Cevik, S., & Jalles, J. T. (2022). This changes everything: Climate shocks and sovereign bonds. *Energy Economics*, 107, Article 105856.
- Chen, G., Chen, W., Wang, J., & Zhao, X. (2023). High-temperature exposure risk, corporate performance and pricing efficiency of the stock market. *Accounting & Finance*, 63(3), 3553–3583.
- Chen, S., & Gong, B. (2021). Response and adaptation of agriculture to climate change: Evidence from China. *Journal of Development Economics*, 148, Article 102557.
- Choi, I. (2001). Unit root tests for panel data. *Journal of International Money and Finance*, 20(2), 249–272.
- Corradin, S., & Schwaab, B. (2023). Euro area sovereign bond risk premia before and during the Covid-19 pandemic. *European Economic Review*, 153, Article 104402.
- Dell, M., Jones, B. F., & Olken, B. A. (2012). Temperature shocks and economic growth: Evidence from the last half century. *American Economic Journal: Macroeconomics*, 4(3), 66–95.
- Deschenes, O., & Moretti, E. (2009). Extreme weather events, mortality, and migration. *The Review of Economics and Statistics*, 91(4), 659–681.
- Deutsch, C. A., Tewksbury, J. J., Huey, R. B., Sheldon, K. S., Ghalambor, C. K., Haak, D. C., & Martin, P. R. (2008). Impacts of climate warming on terrestrial ectotherms across latitude. *Proceedings of the National Academy of Sciences*, 105(18), 6668–6672.
- Di Tommaso, C., Foglia, M., & Pacelli, V. (2023). The impact and the contagion effect of natural disasters on sovereign credit risk. An empirical investigation. *International Review of Financial Analysis*, 87, Article 102578.
- Diarra, I., & Jaber, A. (2022). *Sovereign default risk and climate change: Is it hot enough? SSRN Working Papers, w4193896*. Available at SSRN: <https://ssrn.com/abstract=4193896>. doi: 10.2139/ssrn.4193896.
- Donat, M. G., Alexander, L. V., Yang, H., Durre, I., Vose, R., & Caesar, J. (2013a). Global land-based datasets for monitoring climatic extremes. *Bulletin of the American Meteorological Society*, 94(7), 997–1006.
- Donat, M. G., Alexander, L. V., Yang, H., Durre, I., Vose, R., Dunn, R. J., & Kitching, S. (2013b). Updated analyses of temperature and precipitation extreme indices since the beginning of the twentieth century: The HadEX2 dataset. *Journal of Geophysical Research: Atmospheres*, 118(5), 2098–2118.
- Dunn, R. J., Alexander, L. V., Donat, M. G., Zhang, X., Bador, M., Herold, N., & Bin Hj Yussuf, M. N. A. (2020). Development of an updated global land in situ-based data set of temperature and precipitation extremes: HadEX3. *Journal of Geophysical Research: Atmospheres*, 125(16), Article e2019JD032263.
- Economist Intelligence Unit (EIU). (2015). *The cost of inaction: recognising the value at risk from climate change*.
- Ghaderi, M., Kilic, M., & Seo, S. B. (2022). Learning, slowly unfolding disasters, and asset prices. *Journal of Financial Economics*, 143(1), 527–549.
- Giglio, S., Maggiori, M., Rao, K., Stroebel, J., & Weber, A. (2021). Climate change and long-run discount rates: Evidence from real estate. *The Review of Financial Studies*, 34(8), 3527–3571.
- Graff Zivin, J., & Neidell, M. (2014). Temperature and the allocation of time: Implications for climate change. *Journal of Labor Economics*, 32(1), 1–26.
- Gruber, J. W., & Kamin, S. B. (2012). Fiscal positions and government bond yields in OECD countries. *Journal of Money, Credit and Banking*, 44(8), 1563–1587.
- Heilmann, K., Kahn, M. E., & Tang, C. K. (2021). The urban crime and heat gradient in high and low poverty areas. *Journal of Public Economics*, 197, Article 104408.
- Hong, H., Li, F. W., & Xu, J. (2019). Climate risks and market efficiency. *Journal of Econometrics*, 208(1), 265–281.
- Huynh, T. D., Nguyen, T. H., & Truong, C. (2020). Climate risk: The price of drought. *Journal of Corporate Finance*, 65, Article 101750.
- Jacob, B., Lefgren, L., & Moretti, E. (2007). The dynamics of criminal behavior: Evidence from weather shocks. *Journal of Human Resources*, 42(3), 489–527.
- Jahjah, S., Wei, B., & Yue, V. Z. (2013). Exchange rate policy and sovereign bond spreads in developing countries. *Journal of Money, Credit and Banking*, 45(7), 1275–1300.
- Karydas, C., & Xepapadeas, A. (2022). Climate change financial risks: Implications for asset pricing and interest rates. *Journal of Financial Stability*, 63, Article 101061.
- Kling, G., Lo, Y. C., Murinde, V., & Volz, U. (2018). *Climate vulnerability and the cost of debt*. Available at SSRN 3198093, 2018.
- Klomp, J. (2014). Financial fragility and natural disasters: An empirical analysis. *Journal of Financial Stability*, 13, 180–192.
- Klomp, J. (2015). Sovereign risk and natural disasters in emerging markets. *Emerging Markets Finance and Trade*, 51, 1326–1341.

- Klusak, P., Agarwala, M., Burke, M., Kraemer, M., & Mohaddes, K. (2023). Rising temperatures, falling ratings: The effect of climate change on sovereign creditworthiness. *Management Science*, 69(12), 7468–7491.
- Larraín, G., Reisen, H., & Von Maltzan, J. (1997). *Emerging market risk and sovereign credit ratings*. OECD Development Centre Working Papers.
- Lucas, R. E., Jr. (1978). Asset prices in an exchange economy. *Econometrica: Journal of the Econometric Society*, 1429–1445.
- Maddala, G. S., & Wu, S. (1999). A comparative study of unit root tests with panel data and a new simple test. *Oxford Bulletin of Economics and Statistics*, 61(S1), 631–652.
- Mallucci, E. (2022). Natural disasters, climate change, and sovereign risk. *Journal of International Economics*, 139, Article 103672.
- Nickell, S. (1981). Biases in dynamic models with fixed effects. *Econometrica: Journal of the Econometric Society*, 1417–1426.
- Noth, F., & Schüwer, U. (2023). Natural disasters and bank stability: Evidence from the US financial system. *Journal of Environmental Economics and Management*, 119, Article 102792.
- Painter, M. (2020). An inconvenient cost: The effects of climate change on municipal bonds. *Journal of Financial Economics*, 135(2), 468–482.
- Pankratz, N., Bauer, R., & Derwall, J. (2023). Climate change, firm performance, and investor surprises. *Management Science*, 69(12), 7352–7398.
- Pankratz, N. M., & Schiller, C. M. (2024). Climate change and adaptation in global supply-chain networks. *The Review of Financial Studies*, 37(6), 1729–1777.
- Pástor, L., Stambaugh, R. F., & Taylor, L. A. (2021). Sustainable investing in equilibrium. *Journal of Financial Economics*, 142(2), 550–571.
- Price, J., Warren, R., McDougall, A., VanDerWal, J., Cornelius, S., Sohl, H., et al. (2018). *Wildlife in a warming world: the effects of climate change on biodiversity in WWF's Priority Places*.
- Peppel-Srebrny, J. (2021). Not all government budget deficits are created equal: Evidence from advanced economies' sovereign bond markets. *Journal of International Money and Finance*, 118, Article 102460.
- Sankar, N. G., Nag, S., Chakrabarty, S. P., & Basu, S. (2024). The carbon premium: Correlation or causality? Evidence from S&P 500 companies. *Energy Economics*, 134, Article 107635.
- Somanathan, E., Somanathan, R., Sudarshan, A., & Tewari, M. (2021). The impact of temperature on productivity and labor supply: Evidence from Indian manufacturing. *Journal of Political Economy*, 129(6), 1797–1827.
- Tiihonen, J., Räsänen, P., & Hakko, H. (1997). Seasonal variation in the occurrence of Homicide in Finland[J]. *American Journal of Psychiatry*, 154(12), 1711–1714.
- Yang, L., & Hamori, S. (2023). Modeling the global sovereign credit network under climate change. *International Review of Financial Analysis*, 87, Article 102618.